

## eWater System Overview

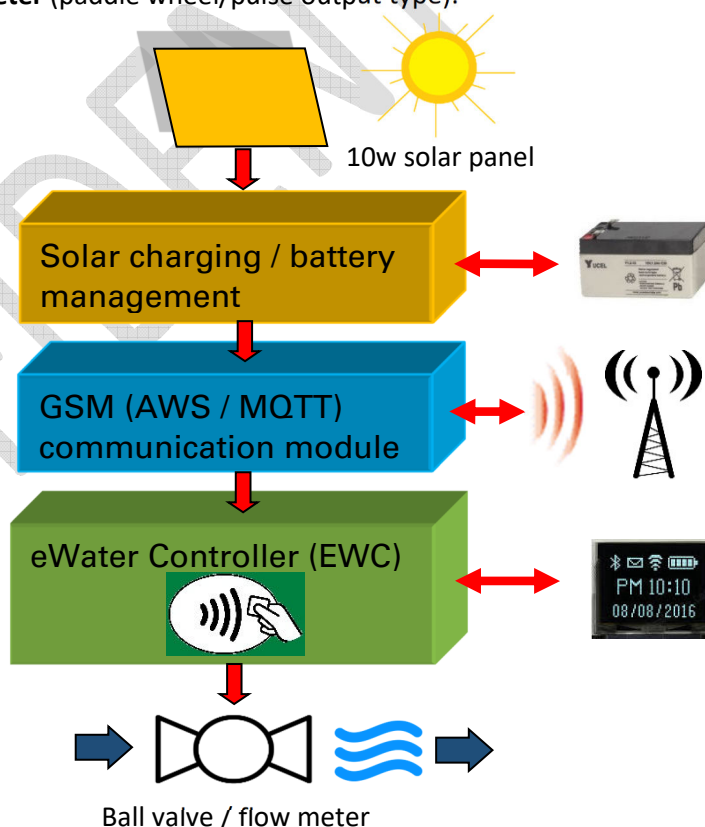
08/03/24 (MITs)

In response to providing sustainable clean water supplies to rural communities in developing countries, “eWater Pay” have designed and manufacture a pre-payment water meter system for fitting to existing water supply standpipes. The water meter is operated by means of an RFID contactless tag where credit is deducted as water is dispensed. The water supply is paid for by pre-charging the tag with credit and the resulting revenue facilitates a business plan to encourage responsibility for the clean water supply and provide employment for “Water Management” and maintenance. Credit and water usage, system events and data logging information are stored locally and transmitted to Cloud services maintaining IoT (Internet of Things) connectivity and control.

## System Description

The pre-payment meter system comprises of the following sub-assemblies:

- 1) **Solar battery charging system** (providing power supply).
- 2) **GSM/GPRS module** (providing AWS / MQTT communication for IoT connectivity).
- 3) **eWater Controller (EWC)** board (NFC Reader and Control system for pre-payment meter)
- 4) **Motorised ball-valve and flow meter** (paddle wheel/pulse output type).



The Solar battery charging system provides power to a GSM module and the eWater Controller board (EWC). The EWC comprises a bespoke NFC Reader and control system that reads credit values from a contactless (NFC) tag and operates a motorised ball-valve whilst measuring the water flow and deducting credit from the tag accordingly. The EWC stores user activity (tag ID and credit values) and logs other data in non-volatile memory. The GSM module supports AWS/MQTT communication and transmits the data log packets to Cloud services and allows system configuration to be monitored and set remotely.

## Solar charging board

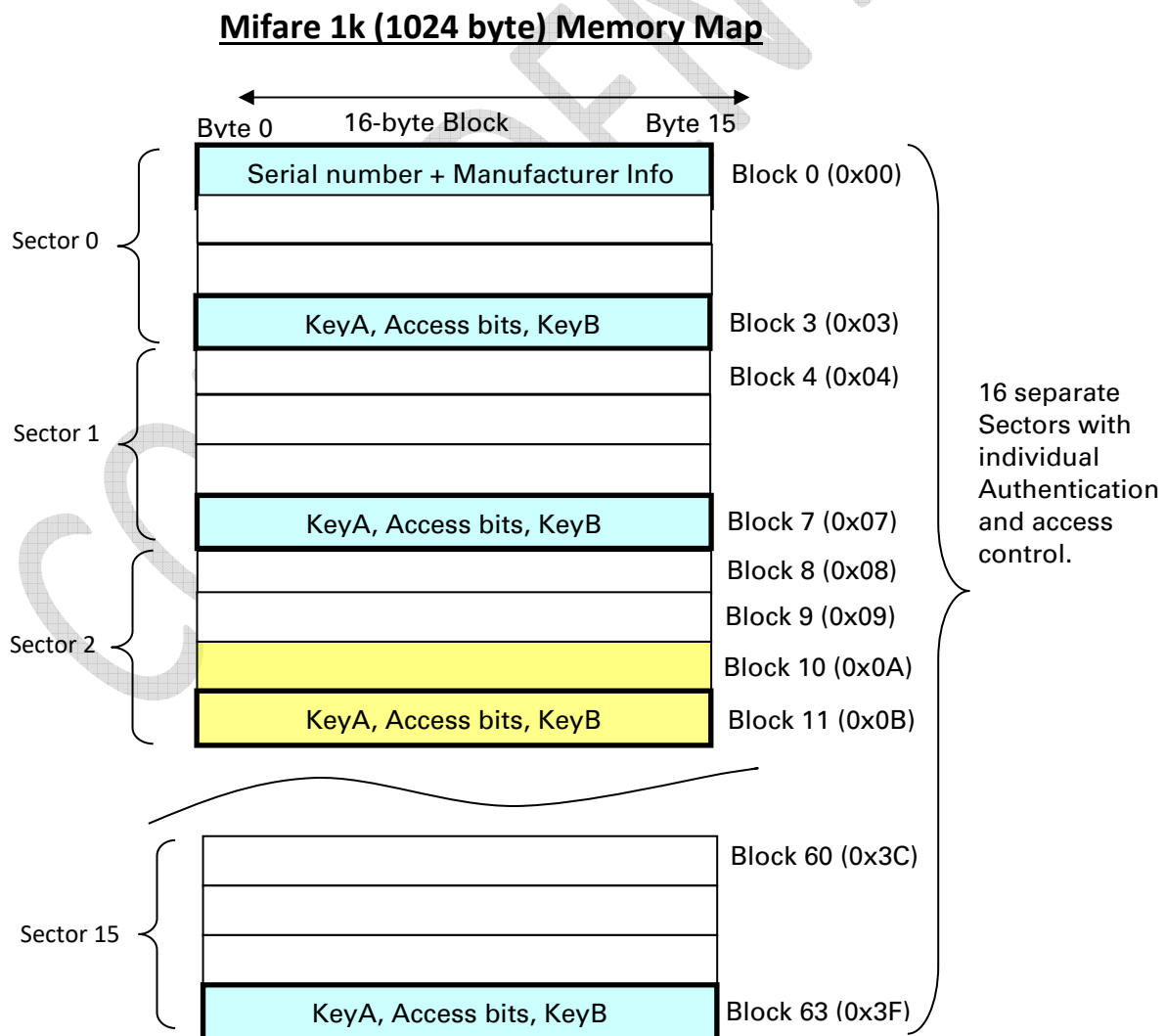
The Solar boards implements a “Linear Technology” LT3652 reference design to charge and maintain a 12v 7Ah SLA battery from a 10w solar panel. The 12v battery provides power to the GSM and EWC boards

## GSM module

The GSM board (project name, Gadwall) uses a Quectel M95 Quad band 2G GSM/GPRS module based on an MTK (MediaTek) chipset. An additional on-board microcontroller provides AWS IoT protocol handling, and power management. The EWC is connected via RS232 serial to the GSM module and implements a proprietary command/response and datalog message protocol. The communication through the Cloud/GSM cellular network is via the AWS/MQTT protocol and the GSM module effectively handles EWC messages and command/responses data transparently with minimum of buffering and latency.

## eWater Controller (EWC)

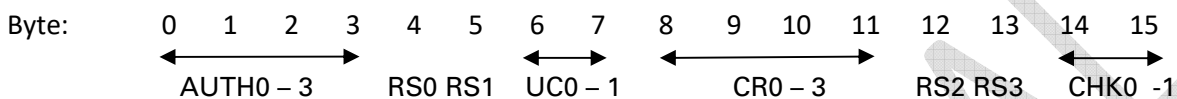
The eWater Controller (EWC) implements an RFID Reader for 13.56MHz ISO14443A Mifare1k “Classic” (NXP S50 chip) cards and tags, this is a sub-set of NFC type 2. These cards have read/write memory allocated as 16-byte BLOCKS. These Blocks are grouped into SECTORS containing 4 blocks each.



Each SECTOR can have read/write access controlled by configuration bits and KEYCODES (passwords) defined in the last block of each sector. This access control method and the associated hardware-based encryption is called the CRYPTO1 system. The Mifare memory structure and the CRYPTO1 hardware access control system is proprietary to NXP Semiconductors (formerly Philips Semiconductors) and is defined and documented in their data sheets and application notes.

The EWC Reader design uses an NXP RF chip and associated microcontroller firmware to provide the CRYPTO1 support to allow read/write memory access to the Mifare cards. **This built-in “hardware” access control system provides the first level of protection.**

The EWC system uses Block 10 of the Mifare card to store the various parameters as shown below:  
Block 9:



### **AUTH0 AUTH1 AUTH2 AUTH3**

AUTH0 – 3 is a four-byte code that will authenticate the cards as eWater card types only so preventing other cards and tags from being used.

The eWater authentication code (AUTH0 - 3) is also stored in the EWC non-volatile EEPROM memory to allow the card to be authenticated as a valid card when presented.

Default value: 0x8F 67 CE B9

### **RS0 RS1 RS2 RS3**

RS0 -3 are reserved bytes for future use (default values RS0 - 1 = 0x00 00, RS2 - 3 = 0xFF FF)

### **UC0 UC1**

UC0-1 is a 16-bit usage up-counter. The usage counter increments under the EWC software control when a card is first presented and authenticated and is written back to the card. This is used as an anti-cloning technique, so the encrypted data cannot be read and written back to the same card without credit being purchased. The usage counter is only “zeroed” when credit is purchased “officially”.

The usage counter should therefore always appear higher and if less would imply data has been copied and written back to the same card. This would allow a card to be “blocked” or “black listed” from the system.

### **CR0 CR1 CR2 CR3**

CR0 – 3 is the credit value in MilliCredit (MITs) units. It is decremented by the EWC system as water is dispensed until a zero value is reached. Further credit (Litres) will then have to be purchased.

### **CHK0 CHK1**

CHK 0 – 1 is a 16-bit checksum of preceding bytes 0 – 13 of the block 10 data. This is used to ensure data integrity in case of memory failures (maximum write cycles have been exceeded) or data ripping issues.

Card Block 9 structure (default values, with UC = 0, CR = 0, checksum = 0x047B)

Byte: 0    1    2    3    4    5    6    7    8    9    10    11    12    13    14    15

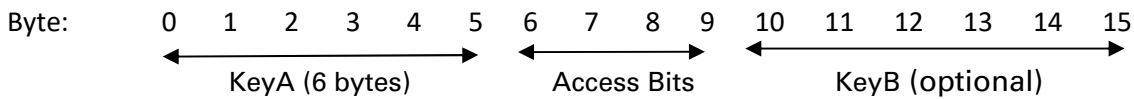
      0x8F 67 CE B9 00 00 00 00 00 00 00 00 FF FF 04 7B

(arbitrary Hex bytes used as security codes so no clues or links to predictable ASCII strings or sequences)

Block 10 (0x0A) is in memory SECTOR 3.

Access control to SECTOR 3 is provided by the KEYCODE and Access Control bits defined in BLOCK 11 (0x0B).

### **Block 11 (0x0B) Sector Trailer Block**



Block 11 Sector Trailer block contains the KeyA and KeyB keycodes (passwords) used to unlock the memory during EWC read/write operations. Block 10 is therefore effectively protected and locked for access by the Block 11 (0x0B) Sector Trailer Block keycodes

Sector Trailer Block 11 (0x0B) Key A = "0xEF F2 6D 53 42 8C", key B = 0x81 CD 94 EC AA 74 as default (but are stored in EWC EEPROM memory and can be changed).

Note Key B not used but defined anyway as arbitrary bytes to avoid any predictable sequences

### **Block 11 (Sector Trailer Block) default values**

|       |      |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|-------|------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Byte: | 0    | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 |
|       | 0xEF | F2 | 6D | 53 | 42 | 8C | FF | 07 | 80 | 69 | 81 | CD | 94 | EC | AA | 74 |

Bytes 6 – 9 0xFF 07 80 69 are the default Access Control bits as defined by NXP. These will remain at default settings for now.

**As additional protection to the data stored on the Mifare cards, the EWC firmware implements a "Dynamic" 32-bit software encryption system that uses the card UID (Unique Identifier or Serial number) as one of the seeds to an encryption algorithm that effectively "scrambles" the data stored on the card such that even the same data stored on multiple cards would always appear different. Additional encryption seeds are stored in the EWC no-volatile memory and allow the encryption algorithm to be changed remotely if compromised.**

The data must therefore be stored on the Mifare card BLOCK 10 in the correct "encrypted" format such that the EWC Reader can corresponding decrypt the data into sensible information that is subsequently processed.

### **EWC2 hardware and operation**

The EWC implements a PIC16F887 microcontroller-based RFID reader design using NXP MFRC531/CLRC632 RF chip with 4-loop PCB antenna to communicate with Mifare cards and tags. Low-power LT1383 provides external RS232 (Tx, Rx and CTS) connection for host communication (GSM module to/from Cloud). PIC software communicates with peripheral devices via SPI bus. Specifically, NXP MFRC531 (SPI RFID transceiver chip) as described above, Microchip 25LC256 (SPI 32k byte EEPROM) and NXP PCF2123 (SPI RTC chip + 32k crystal + optional battery back-up). A directly interfaced BD6221F chip provides H-bridge drive to a motorised ball-valve. Flow meter hall-effect pulses are counted via PIC hardware counter. The principle of operation for EWC is to operate in sleep mode with brief wake-up periods to process RFID activity, control the valve and maintain host communication. The flow meter pulse counter can continue during sleep. A reed switch and a magnet in the RFID tag provide additional "proximity detection" for power control. The extensive use of low-power software techniques with variable sleep periods ensures minimum average power consumption whilst maintaining an active real-time response system.

EWC2.5 upgrade replaces 25LC256 with larger AT25M02 (256k byte) SPI EEPROM to provide storage for up to 8192 datalog packets. An in-line pressure "micro-switch" (set to about 2 PSI / 0.13 bar) is connected to a PIC input so software can detect if no water pressure is available and react accordingly.

## Auxiliary connectors and soft-UART pinout definitions

The EWC has three soft-UART communications channels defined:

COM1: PL1 RS232-level serial Tx, Rx, CTS for GSM (Gadwall) module communication.

COM2: PL2 TTL-level serial Tx, Rx and CTS for auxiliary communication to Display-Lite OLED module.

COM3: CN1 TTL-level serial Tx only for auxiliary communication to MicroSD bulk storage module.

All have the serial protocol defined as 9600 baud, 8-bits, 1-stop, no-parity with CTS hardware handshaking if Rx, receive channel is implemented.

NOTE: COM1 is RS232-level (+/- 10v differential) signals. COM2 and COM3 are 5v TTL level.

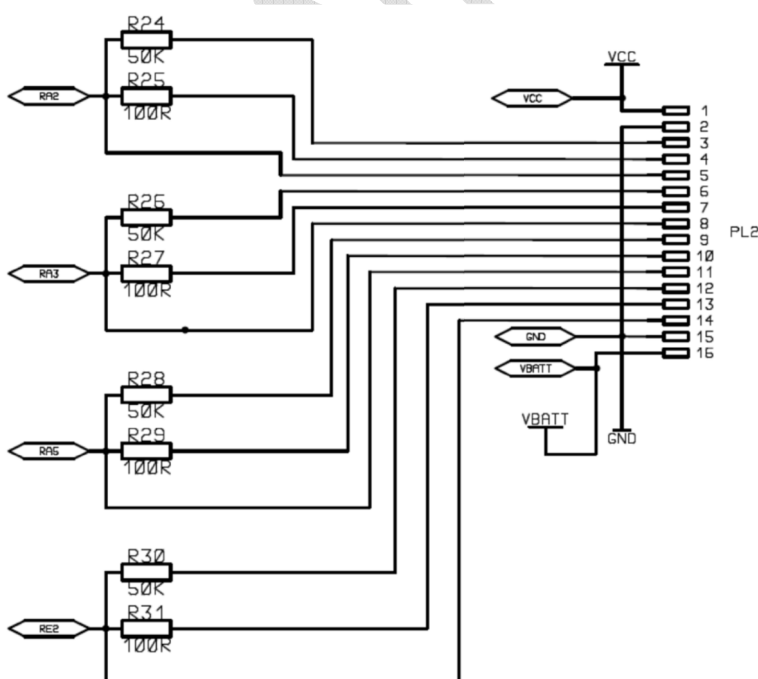
### COM1 (soft-UART1)

The main EWC communications PL1 port to the host device (Gadwall GSM) is a soft-UART defined as COM1 with PIC microcontroller I/O pins used as Tx, Tx and CTS signals. These signals are connected to an RS232 converter so COM1 is a true (+/- 10v differential) RS232 level serial port with Tx, Rx signals and CTS being used for hardware handshaking. COM1 is typically used by the host device to send commands to the EWC and receive replies and error/event datalog packets.

### COM2 (soft-UART2)

The EWC has 4 x "spare" microcontroller I/O pins connected to the Auxiliary port PL2 'Black' connector. These connections also have multiple configurations with both direct and series resistor options. This allow the I/O pins to be used for multiple software-defined purposes including ADC inputs with potential dividers and digital I/O with direct connection or with series 100R (for noise reduction) or 50k (pull-up/down) resistor connections. The various connection and series resistor options can be configured by connecting different pinout combinations.

For the current EWC firmware version a soft-UART (COM2) has been defined so the PIC microcontroller I/O pins are used as TTL level Tx, Tx and CTS signals. This is typically used to output SMARTD commands and data packets to the Display-Lite OLED module (if fitted).



Typical FTDI TTL CABLE  
SERIAL-to-USB wire colours  
For diagnostic purposes

- 1: EWC\_VCC (5v)
- 2: EWC\_GND
- 5: TTL-level Tx (o/p)
- 8: TTL-level Rx (i/p)
- 11: TTL-level CTS (o/p)
- 14: PWR\_EN (o/p)
- 15: EWC\_GND
- 16: VBAT (12v)

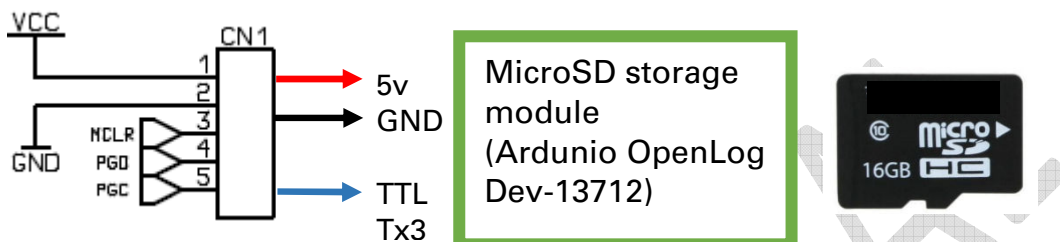
- VCC (only connect 5v if EWC 12v NOT used)
- GND
- Rx (i/p) yellow
- Tx (o/p) orange
- CTS (i/p) brown

The EWC\_VCC (5v), VBAT(12v) connections can be used to power appropriate external devices.

PL2 (BLACK CONNECTOR) COM2 "Auxiliary" port Pinout typically used for interfacing Display-Lite OLED module.

### **COM3 (soft-UART3)**

In addition, the 5-way CN1 header pins on the EWC PCB are defined as both programming pins (5v, GND, MCLR, PGD, PGC) and also I/O pins with specifically PGD (RB7, PGD pin as Tx3 o/p) now used to output the full 39-byte datalog packet in real-time (9600 baud, 8-bits, 1-stop, no-parity). This effectively duplicates the datalog packet transmission via the main COM1 channel to the Gadwall GSM board. An external MicroSD storage module can be connected to CN1 (Ardunio "OpenLog" MicroSD storage module for example) which captures all the datalog packets sequentially in a FAT32 file (on the MicroSD card). This provides a bulk store facility and backup of large amounts of EWC data.



### **CN2 (White) connector pinout**

| Pin1                    |        |         |                       |       |              |      |      | Pin 9        |
|-------------------------|--------|---------|-----------------------|-------|--------------|------|------|--------------|
| HE input,               | HE 5v, | HE GND, | OUT1,                 | OUT2, | Tamp2,       | GND, | GND, | VWAT         |
| (Hall-effect pulse i/p) |        |         | (12v Valve drive o/p) |       | (Switch i/p) |      |      | (ADC1 input) |
| ( Meter cable           |        | )       | ( Valve cable )       |       |              |      |      |              |
| (Yellow                 | Red    | Black ) | ( Black               | Red ) |              |      |      |              |



## EWC2 LED colours and meaning



### **BLUE LED**

**STANDBY HEARTBEAT.** Brief flash every second indicates tap is powered-up and operational. The period between flashes increases to 2-seconds after an hour and then 4-seconds after 2-hours of inactivity.

**TAG WRITE.** When dispensing the blue LED also flashes briefly each time the tag credit is rewritten. This flash during dispensing indicates the flow meter is operational and that flow pulses are being acquired correctly.

### **PROX (MAGNET) DETECT**

When a magnet is presented to the tap (no tag), the blue LED flashes 3-times and the PROX detect message is transmitted (and stored).



### **RED LED**

**FAULT.** Solid red LED indicates a fault condition with the tap or more usually when a tag is presented that is not recognised (not formatted or damaged).

### **STANDBY HEARTBEAT (if NO water pressure detected)**

Brief flash of Red LED every second indicates tap is powered-up and operational BUT pressure switch is open indicating no water pressure (< 2 PSI / 0.13 bar). This can occur in conjunction with other LED indications.

## **YELLOW LED**



**NO CREDIT.** Solid yellow LED indicates a tag has been presented with zero credit. During dispensing, when credit has reached zero, the yellow LED will be turned ON.

**GSM NETWORK STATUS.** Brief flash of the yellow Led during standby indicates the tap is NOT connected to the network (via GSM).

## **GREEN LED**



**WATER DISPENSING.** Solid green LED indicates the valve is open and water is being dispensed. When the tag or tap credit is below the defined low-limit, the green LED will toggle on/off every 0.5 seconds during dispensing as a warning.

**FACTORY RESET:** Green LED toggles every 2.5 seconds during 80+ second memory clearing operation.

## **COMBINATIONS OF LEDs**

### **GREEN + RED**



Solid green and red LED during start of dispensing indicates low-flow detected. The valve will turn OFF and the tap will lock for 30-seconds (or until tag removed) and then enter standby mode again.

### **GREEN + YELLOW**



Solid green and brief yellow LED flash during dispensing indicates the tap is NOT locked on the network (via GSM). The tap operates normally, and dispensing transactions are stored locally in the tap independent of network connectivity.

### **GREEN + BLUE**



Solid or sequences of rapid flashing green and blue indicates a tap hardware fault has been detected. Rarely seen but typically as a result of water ingress to the electronics.

### **RED + BLUE**



Brief flash of the Red and blue LEDs every 8-seconds indicates that low-battery voltage has been detected. The tap remains responsive to system commands but will not dispense water. When the battery voltage increases to above the defined limit, the tap enters standby mode again and normal operation can resume.

### **GREEN + YELLOW + RED + BLUE**



All LEDs flashing for a period of a few seconds indicates tap power-up. Repeated periods of all LEDs flashing indicate that battery voltage is very low and that brown-out of the electronics is occurring. This most likely signifies a faulty or very flat battery.

### **GREEN + YELLOW + RED + BLUE**



LEDs cycle and turn on/off in sequence for a period of a few seconds. This "walking" LED pattern occurs when a tag is presented and a SuperTap credit top-up has occurred.

A single cycle of the LEDs, blue -> red -> yellow -> green indicates that the tag has been formatted to a zero-credit default state (following a FORMAT\_ID\_FAIL or FORMAT\_CHKSUM\_FAIL error). The tag can then be registered on the system and topped-up with credit as required.



## EWC2 RS232 commands and configuration parameters

The EWC (eWATER Controller) has an RS232 serial communication port with Tx, Rx and CTS connections, supporting a Command/Response protocol and a separate automatic data output message protocol. The EWC operates by having brief processing periods (to handle RFID and meter control operations) between low-power “sleep” periods of up to 8-seconds. The CTS signal (an output from the EWC) is used by the GSM module to manage hardware-handshaking so a command sequence is only passed to the EWC when the CTS signal goes low, and the EWC subsequently replies with an acknowledge byte and data. The Command/response protocol is based on defined header bytes and ETX trailing bytes with XOR checksum, transmitted and received as serial data, 8-bit HEX bytes at 9600 baud, 8-bits, 1-stop, no-parity.

Timing wise, the CTS handshaking signal goes HIGH -> LOW for about 15mS every “awake” cycle. These occur every second when idling with tap OFF, every 0.5 seconds when tap is ON. The idling sleep periods increase to 2-seconds after 1 hour and 4-seconds after 2-hours if no activity. In low-battery conditions the sleep periods extend to 8-seconds. The reply to a command occurs after the CTS goes LOW and the automatic datalog transmission is the last operation before the EWC sleeps again.

When the EWC receives a command, the “sleep” period is skipped and the unit polls quickly for a timeout period. This means that the first command can take up to 8-seconds to be processed when CTS goes low (because of the sleep periods) but subsequent commands can be sent, and replies received quickly.

The “Automatic DATALOG output” feature differs in that it is transmitted to the host automatically after an “event” or “error” and no reply is expected. When enabled, the automatic DATALOG packet is transmitted in real-time at the same time the DATALOG information is stored in the SPI EEPROM.

### Typical command sequence:

**<CMD> <ADR> <ARGUMENTS/DATA n> <ARGUMENTS/DATA n+1> ..... <ETX> <XOR>**

<CMD> = command code. For example, PROGRAM EEPROM (ASCII "P", 0x50)

<ADR> = ADDRESS parameter (only included for “addressed” EEPROM reads/writes etc)

<ARGUMENTS/DATA> = argument and data bytes (specific to the command)

<ETX> = 0x03 end of sequence byte

<XOR> = 8-bit XOR of all bytes up to and including ETX.

Example, program EEPROM byte 24 (0x18) = 01

Command sequence (sent when CTS goes LOW) = 50 18 01 03 34

### Typical reply sequence:

**<ACK/STATUS> <CMD> <ADR> <DATA n> <DATA n+1>... <ETX> <XOR>**

<ACK/STATUS> = Acknowledge/status byte. Typically, 0x80 (or 0x88 with RS232 error)

<ADR> = Address parameter if specified in the command sequence.

<DATA> = data bytes returned (specific to the command and may include the ADDRESS parameter)

<ETX> = 0x03 end of sequence byte

<XOR> = 8-bit XOR of all bytes up to and including ETX.

Example, reply to program EEPROM command sequence above = 80 50 18 03 CB (no errors)

Or reply indicating RS232 error = 88 50 18 03 C3

### **RS232 error** occurs if:

Command code is unknown.

Number of arguments is incorrect for the specific command.

No ETX at end of the string.

Calculated XOR value is incorrect.

**Internal EEPROM error** is a fatal h/w error (very unlikely)

The status flags returned in the Acknowledge byte:

b7 b6 b5 b4 b3 b2 b1 b0

1 0 0 0 1 0 0 1

↓ Internal EEPROM error  
RS232 (Protocol) error

## Automatic DATALOG output feature

### Error/Event codes

|              |                                                                                      |
|--------------|--------------------------------------------------------------------------------------|
| D'0'         | No Error                                                                             |
| D'1'         | No Credit (CR = 0)                                                                   |
| D'2'         | Format event due to tag block ID bytes wrong (no match)                              |
| D'3'         | Not MIFARE1k card                                                                    |
| D'4'         | Keycode loading error                                                                |
| D'5'         | Card communication error                                                             |
| D'6'         | Mifare Authentication error (CRYPTO1 error)                                          |
| D'7'         | Format event due to tag block checksum wrong (corruption)                            |
| D'8'         | EEPROM WRITE error (internal)                                                        |
| D'9'         | No card error (not detected by sensor/no RF)                                         |
| D'10' (0x0A) | RS232 CMD error (unknown CMD or arguments/checksum wrong)                            |
| D'11' (0x0B) | Dispense Limit error/event (flow count overflow or max time exceeded)                |
| D'12' (0x0C) | EEPROM write error (external)                                                        |
| D'13' (0x0D) | MFRC chip error                                                                      |
| D'14' (0x0E) | Block Read error                                                                     |
| D'15' (0x0F) | Block Write error                                                                    |
| D'16' (0x10) | NO_FLOW error (typically flow volume < 1 litre after time parameter period)          |
| D'17' (0x11) | PROX event (Magnet detected BUT no tag)                                              |
| D'18' (0x12) | LOW-BATTERY event (battery voltage < low voltage limit parameter)                    |
| D'19' (0x13) | PRESSURE event (change in status of water-pressure micro-switch)                     |
| D'20' (0x14) | SuperTap tag/tap TOP-UP complete                                                     |
| D'21' (0x15) | Host VALVE OFF command received and processed (water dispensing terminated)          |
| D'22' (0x16) | START_UP (power-up / reset) event                                                    |
| D'23' (0x17) | NO_FLOW REPEAT error (set if NO_FLOW and tag left in place, cleared by removing tag) |
| D'24' (0x18) | TAMPER event (change in status of Tamper 1 and / or Tamper 2 switches)               |
| D'25' (0x19) | HEALTH_STATE event (periodic system status report)                                   |

The EWC automatically outputs the DATALOG packet in real-time with header, terminator and XOR checksum (after the data is stored in the internal SPI EEPROM). This feature operates separately to the Command/response protocol controlled by the CTS handshaking and occurs after CTS goes HIGH->LOW->HIGH so should not conflict if enabled. No response is expected.

The automatic DATALOG packet is created as 39-bytes of hex data:

<HDR> <EWC\_ID0, EWC\_ID1, EWC\_ID2, EWC\_ID3>, < 28-byte DATALOG packet >, <CONVH, CONVL>, <DLPH, DLPL>, ETX, XOR

<HDR> = Datalog packet header code = ASCII "D", 0x44

<EWC\_ID0 – 3> is the 4-byte EWC\_ID (MSB first),

<28-byte DATALOG packet> = see below

<CONVH, CONVL> = Litre / Credit conversion parameter (0 – 65535 as MSB, LSB hex bytes)

<DLPH, DLPL> = Datalog pointer (0 – 8190 as HIGH, LOW hex bytes). This the storage pointer for the data in the storage EEPROM.

Wraps around to 00 00 when EEPROM is full and DLP = 1F FE (8190 dec) (Note EWC2, 1022 max)

<ETX> = 0x03 end of sequence byte

<XOR> = 8-bit XOR of all bytes up to and including ETX.

### < 28-byte DATALOG packet >

EE SS MM HH DD MT YY UU UU UU UU AN RS UC UC SCR SCR SCR SCR ECR ECR ECR ECR FC FC FC FT FT  
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Where EE = Error/Event code

SS MM HH = Seconds, Minutes, Hours bytes (BCD), DD MT YY = Day, Month, Year bytes (BCD)

UU UU UU UU is the 4-byte card UID (MSB -> LSB)

AN = Battery voltage ADC value (calibration: voltage = ADC value/256 x 15, for example 12v = 0xCC)

RS = Reserved for particular data depending on the specific error/event (default 0x00).

UC UC is the 2-byte usage counter

SCR SCR SCR SCR is the 4-byte start Credit value (MSB -> LSB) stored as MITs (MilliCredits)

ECR ECR ECR ECR is the 4-byte end Credit value

FC FC FC is the 3-byte (MSB, MID, LSB) Flow meter count (flow volume). Litres  $\approx$  FC / 360

FT FT is the 2-byte (MSB, LSB) Flow time (seconds)

This 39-byte hex data packet is transmitted to the host automatically after a "water dispensing" event. No response is expected. The host system will have to check the Command header code, the length of the data, the ETX terminator and the XOR is correct before the various data fields are extracted.

Examples of the 39-byte auto datalog transmission are:

**44 00 00 00 01 09 25 04 12 01 01 15 F5 7E 43 26 CC 00 00 03 00 00 1F 3E 00 00 1F 37 00 08 C5 00 0A 01 4A 00 30 03 31**

Broken down as (Hex bytes grouped into data fields):

**<44> <00 00 00 01> <09> <25 04 12> <01 01 15> <F5 7E 43 26> <CC> <00> <00 03> <00 00 1F 3E> <00 00 1F 37> <00 08 C5> <00 0A> <01 4A> <00 30> <03> <31>**

In this case the error/event code is 0x09 = no card (tag removed)

When a pure magnet is presented (no RFID), the "PROX event" packet has following typical format:

**44 00000001 11 220612 010115 00000000 CC000000 00000000 00000000 000000 0000 014A 0036 03 09**

The 0x11 (17 dec) error/event code shows the "Prox event"

Occasionally, if a magnet is slid into place and the tag is pushed out quickly while water is being dispensed then the Prox event will terminate the water dispensing session and the packet will contain data similar to below.

**44 00000001 11 110512 010115 F55F63A3 CC000006 00001ED2 FFFFFFFF 000000 0001 014A 0033 03 9D**

Note, the FFFFFFFF end credit field indicates that the credit has NOT been written to the tag.

### "SUPERTAP" feature and command/response protocol

The SuperTap feature has two modes of operation, TAG and TAP top-up:

- 1) The EWC can "top-up" or write credit to 25 x user TAGS (using LOAD and CLEAR SuperTap table commands to manage PIC EEPROM list of 25 x UID + TOP-UP values).
- 2) The EWC can use HOST VALVE-ON/OFF command to transfer credit to the tap (without tag) and operate remotely. Subsequent TAP TOP-UP + CREDIT commands can "top-up" the credit on the TAP. This allows tap to be used remotely for "in-house" applications.

### **SuperTap TAP top-up (NOTE: the tap unit itself)**

The EWC can be remotely controlled by host commands (without using RFID tag). The VALVE ON command loads TAP credit and turns the valve ON. As water is dispensed the internal TAP credit is decremented and the valve turns off automatically with a zero-credit datalog event when no credit remains. Before reaching zero credit the TAP TOP-UP command can be used to "top-up" the TAP credit, so water dispensing is not interrupted. The VALVE OFF command can be used at any time to terminate the dispensing. The TAP TOP-UP and VALVE OFF commands generate a datalog event packet in addition to the acknowledge reply to the command.

LOW\_BATT and NO\_FLOW errors can also terminate the TAP dispensing with appropriate datalog event packet stored and transmitted.

#### **"HOST VALVE ON" command**

Turns VALVE ON manually by host control and loads "start credit" value (RFID tag operation is disabled).

**<ASCII "V", 0x56> <SCR> <SCR> <SCR> <SCR> <ETX> <XOR>**

Where <SCR> <SCR> <SCR> <SCR> = start credit value (32-bit integer, MSB byte first)

Example: 0x56 00 00 00 64 03 31

Where start credit (litres required) = 0x00 00 00 64 = 100 dec.

Reply: 0x80 <CMD> <SCR> <SCR> <SCR> <SCR> <ETX> <XOR> if OK.

Where <CMD> = 0x56, <ETX> = 0x03

Where <SCR> <SCR> <SCR> <SCR> = start credit loaded (MSB first).

(or 0x88 56 03 DD error reply if VALVE already ON, LOW-FLOW lock-out, LOW-BATT or number of arguments, ETX, XOR wrong)

#### **"HOST VALVE OFF" command**

Turns VALVE OFF manually by host control and returns end (remaining) credit value.

Datalog packet is also stored and transmitted (with VALVE OFF event code).

RFID operation is enabled afterwards.

**<ASCII "O", 0x4F> <ETX> <XOR>**

Example: 0x4F 03 4C

Reply: 0x80 <CMD> <ECR> <ECR> <ECR> <ECR> <ETX> <XOR> if OK

Where <ECR> <ECR> <ECR> <ECR> = end (remaining) credit value (MSB first) and valve OFF.

VALVE\_OFF event datalog packet generated.

0x88 4F 03 C4 error reply if number of arguments, ETX or XOR wrong or If HOST VALVE\_OFF received when RFID enabled.

if end credit value = 0xFF FF FF FF then valve was already OFF (as a result of an error interrupting the water dispensing).

So, in summary VALVE\_OFF returns 80 4F FF FF FF FF 03 CC and NO Datalog packet if valve already OFF.

#### **"TAP top-up" command**

Adds (tops-up) credit to existing VALVE ON credit during dispensing operation.

Datalog packet is also stored and transmitted (with TOP-UP success event code).

**<ASCII "U", 0x55> <SCR> <SCR> <SCR> <SCR> <ETX> <XOR>**

Where <SCR> <SCR> <SCR> <SCR> = Top-up credit value (32-bit int, MSB byte first)

Example: 0x55 00 00 00 64 03 32

Where top-up credit (litres required) = 0x00 00 00 64 = 100 dec.

Reply: 0x80 <CMD> <ECR> <ECR> <ECR> <ECR> <ETX> <XOR> if OK

Where <ECR> <ECR> <ECR> <ECR> = accumulative (remaining) credit value (MSB first) and TOP-UP event datalog packet generated.

88 55 03 DE error reply if number of arguments, ETX, XOR wrong or TAP TOP-UP received when RFID enabled.

A user can "top-up" the VALVE-ON credit before the credit runs out, so the water flow is maintained. LEDs cycle as visual indication and TOP-UP event datalog packet is generated.

**NOTE the "HOST VALVE ON/OFF/TOP-UP" commands use the same start / end credit philosophy as using the Tags to dispense water.**

**When the VALVE ON command is received with a start credit, the water is dispensed and the credit decrements until zero. The valve then turns OFF and a ZERO-CREDIT datalog packet is generated.**

**The packet contents will show the time and start/end credit details for the water dispensed.**

**Various error conditions (low-battery, low-flow etc) can also interrupt the water dispensing, turning the valve OFF and generating corresponding error/event datalog packet.**

**The VALVE OFF command interrupts the water dispensing, turns the valve OFF and returns the remaining (end) credit value AND generates a datalog packet with all the time and start/credit details.**

**Sending VALVE OFF again returns end credit of 0xFF FF FF FF indicating valve already OFF.**

NOTE that during the VALVE ON dispensing period the EWC will automatically generate a datalog packet when the flow-count reaches a defined limit (EEPROM parameter) or when the dispensing period passes a Dispense-time limit (EEPROM parameter defined).

This will allow the host to know the tap is still functioning and dispensing water during long "Valve ON" periods. The datalog packet in this case will have the "Dispense-Limit error/event code (11 dec, 0x0B) set and the valve will briefly turn OFF/ON (a feature to exercise the valve and clear sediment). If the NO\_FLOW error control parameter in EEPROM is disabled, then the EWC could be used as an in-house valve/meter with flow controlled by a physical ON/OFF tap.

### **SuperTap TAG top-up (NOTE: the NFC / RFID tag)**

The EWC can be used to add (top-up) or write specific credit to a user's tag as well as decrementing credit as water is dispensed. In areas with poor mobile internet (Data services) coverage, the user can "buy" credit using SMS based mobile money (M-Pesa, Tigo-Pesa, Airtel Money etc). The credit purchase is confirmed by SMS and the text message directs the users to go to their local EWC tap to "pick-up" the credit. Practically, the tag ID and top-up credit value are sent to a table in the relevant EWC.

When the tag is presented and UID matched, the EWC adds the top-up credit to the value already on the tag or writes a specific value before water dispensing continues. This is called the "Supertap" feature.

The Supertap UID and top-up credit table is located in non-volatile PIC EEPROM memory (n x (4-bytes + 4-bytes) and can hold 25 entries. Entries are identified as SuperT location 0-24.

**Note the SuperTap table is NOT cleared on power-on/reset.**

The SuperT UID/Top-up values are received as two encrypted 32-bit values. The values are decrypted for storing in the table as plain data. The Encryption/Decryption algorithm is the same as used for the Tag data. For the SuperT data, a separate decryption mask is stored in the EEPROM (as a pseudo UID for seeding the algorithm). The host system uses the same encryption/decryption mask and seeds, so the data matches on the host and EWC.



**EWC SuperT table in EEPROM**

| Loc | UID         | Top-UP                                                           |
|-----|-------------|------------------------------------------------------------------|
| 0   | XX XX XX XX | XX XX XX XX (Values stored MSB -> LSB)                           |
| 1   | XX XX XX XX | XX XX XX XX                                                      |
| 2   | 00 00 00 00 | 00 00 00 00 (zero entry for UID and top-up indicate blank field) |
| ↓   |             |                                                                  |
| 24  | XX XX XX XX | XX XX XX XX                                                      |

The table is managed by the host system and the UID/top-up fields are cleared only by the EWC when top-up credit has been applied to a tag correctly, or by the host clearing the entry. The table is not sorted so blank fields will have to be remembered by the host and a mirror image maintained. When the users tag is presented, if the UID matches a SuperTap entry, the corresponding top-up value is added or written to the tag's existing credit.

**"LOAD SUPERTAP TOP-UP" command**

Loads user UID and TOP-UP credit values into EWC EEPROM memory at specified table location 0-24.

<ASCII "L", 0x4C> <LOC> <UID0> <UID1> <UID2> <UID3>  
<TOP-UP0> <TOP-UP1> <TOP-UP2> <TOP-UP3> <ETX> <XOR>

Where <LOC> = table location 0-24 (as hex byte)

<UID0> <UID1> <UID2> <UID3> = plain/encrypted tag UID (stored MSB first)

<TOP-UP0> <TOP-UP1> <TOP-UP2> <TOP-UP3> = plain/encrypted 32-bit integer "top-up" to add or write to tag (MSB first)

Most Significant bit of <TOP-UP0> indicates WRITE credit (b31 = 1) or ADD credit (b31 = 0)

Maximum credit to WRITE or ADD is therefore 0x7F FF FF FF (2,147,483,647)

NOTE, if enabled (by EEPROM parameter) the SUPERTAP top-up UID and credit value are sent encrypted and decrypted by EWC.

Example (no encryption/decryption):

0x4C 05 04 E0 0F C4 00 01 86 A0

Where UID = 0x04 E0 0F C4

TOP-UP (Addition) = 0x00 01 86 A0 (add 100,000 dec)

Table location = 05

OR

0x4C 05 04 E0 0F C4 80 01 86 A0

Where UID = 0x04 E0 0F C4

TOP-UP (WRITE) = 0x00 01 86 A0 (write 100,000 dec)

Table location = 05

Reply: 0x80 4C <LOC> 03 <XOR>

(or 0x88 4C <LOC> 03 <XOR> error reply if table location is out of range,

NOTE, subtraction using 2's complement addition cannot be used now.



### **"CLEAR SUPERTAP TOP-UP" command**

Clears user UID and TOP-UP credit values in EWC EEPROM memory at specified table location 0-24.

**<ASCII "Z", 0x5A> <LOC> <ETX> <XOR>** (0x5A xx 03 xx)

Where <LOC> = table location 0-24 (as hex byte)

Example: 0x5A 05 03 5C

Table location 05, UID and TOP-UP entries cleared to zero (00 00 00 00, 00 00 00 00 )

Reply: 0x80 5A <LOC> 03 <XOR>

### **"BULK CLEAR SUPERTAP table" command**

Clears entire 25 x 8-byte (200-byte) SuperTap EEPROM table.

**<ASCII "B", 0x42> <ETX> <XOR>** (0x42 03 41)

Reply: 0x 80 42 03 C1

(or 0x88 42 03 C9 error reply if table location is out of range, or number of arguments wrong)

(or 0x81 42 03 C0 error reply indicating **Internal EEPROM error** (fatal h/w error and very unlikely))

Note the SuperT UID and Top-Up table entry is cleared by following events only:

- 1) When CLEAR SUPERTAP TOP-UP command is received
- 2) When BULK CLEAR table command is received (clears all 25 table entries)
- 3) After user tag has been successfully topped-up and the TOP-UP event message has been transmitted. The EWC LEDs also cycle back and forth to visually show success.

**The successful TOP-UP event message has the same format as the DATALOG packets as below:  
39-byte DATALOG packet format (HEX bytes)**

**<HDR> <EWC\_ID0, EWC\_ID1, EWC\_ID2, EWC\_ID3>, < 28-byte DATALOG packet >, <CONVH, CONVL>, <DLPH, DLPL>, ETX, XOR**

**<HDR>** = Datalog packet header code = ASCII "D", 0x44

**<EWC\_ID0 – 3>** is the 4-byte EWC\_ID (MSB first),

**<28-byte DATALOG packet>** = see below

**<CONVH, CONVL>** = Litre / Credit conversion parameter (0 – 65535 as MSB, LSB hex bytes)

**<DLPH, DLPL>** = Datalog pointer (0 – 8190 as HIGH, LOW hex bytes).

This the storage pointer for the datalog packet in the SPI EEPROM.

Wraps around to 00 00 when EEPROM is full and DLP = 1F FE (8190 dec), (Note EWC2, 1022 max)

**<ETX>** = 0x03 end of sequence byte

**<XOR>** = 8-bit XOR of all bytes up to and including ETX.

### **< existing 28-byte DATALOG packet >**

**EE SS MM HH DD MT YY UU UU UU UU AN ST UC UC SCR SCR SCR SCR ECR ECR ECR ECR FC FC FC FT FT**  
**0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27**

With addition of the SuperT location number ST (0-24) stored in Datalog packet RS0 position. This allows the host to know which of the SuperT table locations was used to apply top-up.

The TOP-UP event could occur before water is dispensed (when tag is first presented) or during the water dispensing period (Load TOP-UP command can be received at any time). The DATALOG packet with TOP-UP event code could therefore have minimal data content or terminate the water dispensing with full information in the data fields.

e.g.

44 00000001 14 220612 010115 04E00FC4 CC 18 0001 00000FA0 00001004 000000 0000 014A 0036 03 81

Where 0x14 (20 decimal) is the TOP-UP complete event code.

SuperT table location used for top-up, ST = 0x18 = 24 dec

Start credit = 0x0FA0 = 8000 dec, End Credit = 0x1004 = 8100 dec

Note End Credit is higher and shows the top-up was 100 decimal.

Note the top-up integer M.S bit flag (for WRITE or ADDITION) is cleared and not stored.

**NOTE** the EWC generally processes and replies to a command during the same cycle and returns to “sleep” when the action and response has been completed. Any errors or events would be detected and processed on subsequent “wake” cycles. This means that repeated commands being sent to the EWC for processing and reply could delay the response to a tag operation and error / event datalog packet transmission / storage.

An exception to this general rule is the SuperTap TAG or TAP top-up commands. These can be acknowledged and the event datalog packet transmitted / stored during the same “wake” cycle. This means the transmitted data packet can be longer than usual with the command reply packet being immediately followed by the datalog packet.

Eg. For example, The TAP top-up command is processed, acknowledged and the datalog packet occurs in the same wake period so the transmitted data is 47 bytes

EWC transmits: 0x80 55 xx xx xx xx 03 XOR 0x44 <EWC\_ID> <28-bytes> CN CN DP DP 03 XOR = 47-bytes

### **PRESSURE error/event datalog packet**

The PRESSURE error/event datalog packet is generated when there is a change in the status of the water-pressure microswitch. The switch is connected to the EWC Vwater ADC input and closing the switch connects the input to ground (zero reading, pressure OK). For NO pressure the input is pulled high and non-zero is returned. The changed status (the Vwater ADC value) is returned in the VW byte of the datalog packet.

#### **< 28-byte PRESSURE DATALOG packet >**

EE SS MM HH DD MT YY UU UU UU UU AN VW UC UC SCR SCR SCR SCR ECR ECR ECR ECR FC FC FC FT FT  
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Where EE = PRESSURE error/event code (D'19')

VW = Pressure-switch status, Vwater ADC reading, zero (Pressure OK), non-zero (NO pressure)

### **TAMPER error/event datalog packet**

The Tamper error/event datalog packet is generated when there is a change in the status of the Tamper 1 (top case) or Tamper 2 switches (bottom case). The switches are NO (normally open) and close when the case fronts are in place. The switches are internally connected to pulled-up inputs and connect to ground when closed. A logic "1" therefore indicates "open". The changed status is returned in the TP byte of the datalog packet.

#### **< 28-byte TAMPER DATALOG packet >**

EE SS MM HH DD MT YY UU UU UU UU AN TP UC UC SCR SCR SCR SCR ECR ECR ECR ECR FC FC FC FT FT  
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Where EE = TAMPER error/event code (D'24')

TP bit allocations

b7 b6 b5 b4 b3 b2 b1 b0

0 0 0 0 0 0 | TAMP2 (Tamper 2 switch, bottom box, open = 1)

TAMP1 (Tamper 1 switch, on solar4 board, open = 1)

### **NO-CREDIT error/event datalog packet**

The datalog packet generated by a NO-CREDIT error/event is a special case where the flow ticks are "sampled" and counted for a 5-second period after the VALVE-OFF occurs. This allows the "un-metered" flow ticks that occur as the motorised ball valve is closing (approx. 3-seconds) to be counted.

In this unique case the un-metered flow tick count overwrites the 2-byte Usage counter in the datalog packet.

#### **< 28-byte NO-CREDIT DATALOG packet >**

EE SS MM HH DD MT YY UU UU UU UU AN RS FCu FCu SCR SCR SCR SCR ECR ECR ECR ECR FC FC FC FT FT  
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Where EE = NO-CREDIT error/event code (D'1')

FCu, FCu = "un-metered" flow tick count MSB, LSB that occurs after VALVE OFF signalled and as ball valve closes.

The FCu flow count can be used to account for all water dispensed and calibrate the tap accordingly.

Practically a 5-second delay occurs when the Yellow LED is lit as the tag/tap runs down to ZERO CREDIT.

### **FORMAT error/event datalog packet**

The datalog packet is generated when an unformatted tag is presented and the EWC has written a default zero-credit block to initialise the tag to a known good state.

The system assumes that the tag has the correct Block 11 Sector Trailer block setting (KeyA and KeyB keycode passwords), if not there is no response or operation, and the RED LED is shown.

If the tag data block ID bytes are wrong then the FORMAT\_ID\_FAIL error is reported, this is typically when the tag is new and unregistered.

If the tag data block Checksum bytes are wrong then the FORMAT\_CHKSUM\_FAIL error is reported, this is typically when the tag data has been corrupted. The datalog packet is stored and transmitted after the tag write has occurred so if no other errors the assumption can be made that the tag has been formatted.

#### **< 28-byte FORMAT event DATALOG packet >**

EE SS MM HH DD MT YY UU UU UU UU AN RS RS RS SCR SCR SCR SCR ECR ECR ECR ECR FC FC FC FT FT  
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Where EE = FORMAT\_ID\_FAIL (error code D'2') or FORMAT\_CHKSUM\_FAIL (error code D'7')  
SCR, ECR, FC, FT fields are zero.

#### **START UP event datalog packet (event code 0x16, 22 decimal)**

On power-on / reset the EWC flashes the LEDs for 7-seconds to allow the host system to synchronise and be ready to receive datalog packets. The START\_UP datalog packet is then transmitted/stored with RTC information and the 4-byte SCRO - 3 field replaced with the power-up count byte (from EEPROM byte 46, 0x2E) and the firmware date, DD MM YY as BCD bytes.

For example:

**44 00000001 16 110512 010115 00000000 00000000 01300118 00000000 000000 0000 014A 0002 03 22**  
where 0x01 is power-up count and 30 01 18 is firmware date as BCD bytes.

## HEALTH STATE event datalog packet

The "Health-state" event datalog packet is generated at specified 1 – 255-minute intervals as defined by the HEALTH\_STATE PERIOD (non-volatile) EEPROM parameter. A zero setting effectively disables the HEALTH\_STATE event so there is no output. Otherwise, the parameter defines the interval between the datalog packet reports up to 255 minutes or 4.25 hours max.

The purpose of the HEALTH\_STATE event is to provide a "push" packet with critical system information and accumulated water usage data in a standard datalog packet format (but with specific and modified data fields). This can be used by the cloud services to confirm the EWC is functioning correctly and to acquire information without requesting it. The benefits are reduced data traffic between the EWC and the cloud services.

**NOTE THAT THE HEALTH\_STATE DATALOG PACKET IS TRANSMITTED BUT NOT STORED IN INTERNAL DATALOG MEMORY.**

The HEALTH\_STATE PACKET is transmitted from the main COM1 (to Gadwall GSM) and COM3 serial ports (to MicroSD bulk storage module, if fitted). The HEALTH\_STATE packet is therefore recorded in the MicroSD bulk storage along with other datalog packets BUT NOT in the INTERNAL datalog memory.

### HEALTH-STATE packet 39-byte structure.

```

<HDR> <EWC_ID0, EWC_ID1, EWC_ID2, EWC_ID3> <EE> <SS MM HH DD MT YY> <UU UU UU UU>
  0      1      2      3      4      5  6  7  8  9 10 11 12 13 14 15
<Vbat Vwat Vsen1 Vsen2 Vsen3> <TACC0, TACC1, TACC2, TACC3, TACC4, TACC5, TACC6, TACC7>
 16 17 18 19 20 21 22 23 24 25 26 27 28
<FLG0> <RS0, RS1, RS2> <CONVH, CONVL> <DLPH, DLPL> <ETX> <XOR>
 29 30 31 32 33 34 35 36 37 38
  
```

<HDR> Datalog packet header code = ASCII "D", 0x44

<EWC\_ID0, EWC\_ID1, EWC\_ID2, EWC\_ID3>, 4-byte EWC\_ID (MSB first)

<EE> Health check event code, 25 decimal, 0x19

<SS MM HH DD MT YY> BCD time and date

<UU UU UU UU> Tag UID if tag is present

<Vbat> Battery voltage, <Vwat> pressure switch/sensor reading

<Vsen1, Vsen2, Vsen3> Sensor voltage readings (if SENSE board fitted)

<TACC0, TACC1, TACC2, TACC3, TACC4, TACC5, TACC6, TACC7>, 8-byte Tick Accumulator

<FLG0>, System flags.

FLG0 bit allocations (FLG1 reserved for future use)

b7 b6 b5 b4 b3 b2 b1 b0

x x x x x x x x

| | | | | | | TAMP2 (Tamper 2 switch, bottom box. NOTE: NOT ON EWC2.3)

| | | | | | | TAMP1 (Tamper 1 switch, on solar4 board)

| | | | | | | GSM\_FLG (1 = GSM not locked, from Gadwall RTS)

| | | | | | | VALVE ON/OFF status (1 = ON, 0 = OFF)

| | | | | | | LOCK\_FLG (1 = no flow lockout timeout in progress)

| | | | | | | PROX\_FLG (1 = Magnet detected but no RFID, event message sent/logged)

| | | | | | | LBATT\_FLG (1 = low battery detected, event message sent/logged)

RFID\_FLG (1 = disabled, manual valve control)

<RS0, RS1, RS2>, Reserved bytes for future use

<CONVH, CONVL>, FCF (ticks per credit conversion factor)

<DLPH, DLPL>, datalog memory pointer

<ETX> = 0x03 end of sequence byte

<XOR> = 8-bit XOR of all bytes up to and including ETX.

## SMARTD consumer data message

The eWater tap supports Display modules.

**DISPLAY-LITE** board. This provides a simple and cost-effective display option using an ALPHANUMERIC OLED display and a small processor-based interface board connected to the COM2 (Black Aux) connector. SMARTD packets can be routed here as required.

When a tag is present, or during manual VALVE-ON operations, if the SMARTD option is enabled (EEPROM parameter) and water is being dispensed, the EWC transmits the SMARTD consumer data message to the Display board when the authorised tag is first detected (or valve-on occurs) and subsequently each time credit is updated (rewritten).

A SMARTD data packet is created that contains the key pieces of information that the SMARTD software can process and display. The SMARTD packet has similar format to the host DATALOG packet with defined header byte, data and with ETX and XOR at end. The SMARTD packet is transmitted to the SMARTD board initially when a tag is presented and subsequently each time a write occurs to the tag. This will enable a user to see real-time information on credit used, litres dispensed etc.

Additional EWC EEPROM stores have been created for the "Litres Conversion Factor", LCF0, 1 and the "Price Conversion Factor", FX0 – FX3 (MSB...LSB 32-bit integer). These are used as conversion constants for the SMARTD board to allow volume and price information to be displayed. The LCF0, 1 and FX0 – FX3 parameters are not used by the EWC and only passed to the SMARTD as part of the packet.

SMARTD packet (all values treated as integers stored MSB first)

**SD < UID0 UID1 UID2 UID3> <CR0 CR1 CR2 CR3> <FC0 FC1 FC2> <FCF0 FCF1> <LCF0 LCF1>  
<FX0 FX1 FX2 FX3> ETX XOR**

Where SD header = ASCII "S", 0x53

<UID0 UID2 UID2 UID3> = Current tag UID

<CR0 - CR3> = Current tag credit MITs (Milli-Credits)

<FC0 FC1 FC2> = Flow ticks

<FCF0 FCF1> = EEPROM "Flow Conversion Factor" (effectively "ticks per credit", default 360 dec)

<LCF0 LCF1> = EEPROM "Litres Conversion Factor" (effectively "ticks per litre, default 360 dec)

<FX0 FX1 FX2 FX3> = EEPROM "Price Conversion Factor". Price as 1/1,000,000 of unit credit

ETX = 0x03

XOR = XOR of all preceding bytes.

The SMART DISPLAY OLED module will show (like a fuel station display but also including tag balance)

|                            |              |                    |              |
|----------------------------|--------------|--------------------|--------------|
| <b>Tag value (Balance)</b> | <b>XXXXX</b> | <b>Price/Litre</b> | <b>XXXXX</b> |
| <b>Transaction cost</b>    | <b>XXXXX</b> | <b>Litres</b>      | <b>XXXXX</b> |

When dispensing, the Tag value (Balance) will decrease, and the cost and litres will proportionally increase. The price/litre is a fixed calculated constant.

Formula used for calculations:

MU (MITs used) = FC x 1000 / FCF

L (Litres) = FC / LCF

V (Tag balance) = MT x FX / 1000

where MT = MITs credit on tag

C (Cost) = FC x FX / FCF

where FX = price per credit (not MIT)

PL (Price/Litre) = FX x LCF / FCF



For example, typical FX (price per credit) in the Gambia = 0.070755 Dalasi / Credit  
 Stored on EWC x 1,000,000 to allow 6 decimal place integer calculations.  
 FX (stored on EWC) = 0.070755 Dalasi x 1,000,000 = 70,755 decimal = 0x00 01 14 63

Scaling to give 3 decimal place display values.

$$V \text{ (Balance)} = MT \times FX / 1,000,000$$

$$PL \text{ (Price/Litre)} = FX \times LCF / FCF \times 1000$$

$$C \text{ (Cost)} = FC \times FX / FCF \times 1000$$

$$L \text{ (Litres)} = FC \times 1000 / LCF$$

For example, if tag credit = 500,000 MITs, FC = 14400, FX = 70755, FCF = 1018, LCF = 360

$$V = 35377 \text{ (35.377 D displayed)}$$

$$PL = 25 \text{ (0.025 D/L displayed)}$$

$$C = 1000 \text{ (1.000 D displayed)}$$

$$L = 40000 \text{ (40.000 L displayed)}$$

Note the "Credit" stored on the tag is in MilliCredit (MIT) units

1000 MITs = 1 Credit. Working with MITs effectively creates 3 decimal place accuracy to the integer maths.

## **Pulse (tick) pre-load concept and implementation**

During water dispensing, the flow meter generates pulses (ticks) that are counted by the microcontroller's internal hardware pulse counter. The pulse rate is directly proportional to the flow rate, so the pulses counted are equivalent to a quantity of water dispensed.

When a tag is presented or a valve ON command is received, the valve ON drive is turned on, the flow meter hall-effect sensor is turned on and the pulse counting begins. The motorised ball valve takes approximately 3-seconds to open. Similarly, when the tag is removed or valve OFF command is received, the valve OFF drive operates and the flow meter hall-effect sensing is turned OFF and the pulse counting stops. However, the valve takes approximately 3-seconds to close and during this time un-metered water continues to flow.

Over time, this repeated un-metered water flow, represents a significant difference in recorded volume versus actual volume dispensed. To counter this issue, a pulse PRE-LOAD concept has been introduced to add "virtual" pulses to the dispensed pulses to compensate for the un-metered and lost revenue water.

### **Pre-load value**

As the motorised ball valve closes over the 3-second period, the bulk of the un-metered flow is during the first 2-seconds of closing, for the remaining one second period, the ball is rapidly throttling the flow until fully closed. So, we can assume that most of the un-metered water occurs over the first 2-second period.

For average flow rates of 30 litres per minute, the flow over a 2-second period is one litre. Therefore, a default pre-load value equivalent to the LCF (ticks per litre value) is a good starting point.

A PRE-LOAD-VALUE EEPROM parameter has been defined so the pre-load value can be altered to match the flow rate.

A PRE-LOAD-PERIOD EEPROM parameter has also been defined as the point at which the PRE-LOAD is added to the dispensing ticks OR the feature can be disabled so no pre-load is applied.

The PRE-LOAD PERIOD parameter can be a tick value 0 – 65534 (0x0000 – 0xFFFF) from the start of dispensing that the pre-load value is added. This occurs only once per dispensing period.

If the PRE-LOAD-PERIOD tick parameter = 0xFFFF (max value) then the pre-load feature is disabled, and no virtual ticks are added.

### Examples of PRE-LOAD operation

The PRE-LOAD-VALUE can be set to any value from 0 to 65535 ticks so the level of pre-load can be altered to match the flow rate.

With the PRE-LOAD-PERIOD parameter set to 0x0000 (zero), the pre-load value is added on the first tag "write" after the tag is presented and dispensing begins. So even for very small, dispensed quantities or if the flow meter was not working (no ticks generated), the tag credit would be immediately deducted by the pre-load equivalent value. This could penalise people who regularly dispense small quantities of water.

With PRE-LOAD-PERIOD parameter set to 0x0168 (360 decimal, LCF value), when the flow ticks exceed 360 then the pre-load value is added to the dispensing tick value.

After the PRE-LOAD-VALUE has been added, the PRE-LOAD-PERIOD value is set to 0xFFFF (65534) as a flag to indicate the pre-load operation is complete.

IF the PRE-LOAD-PERIOD value is too high then it is likely the water dispensing will be finished before the pre-load is added so the un-metered water will not be accounted for.

### PRE-LOAD operation expressed as an algorithm during water dispensing cycle.

Load **PRE-LOAD-PERIOD** parameter from EEPROM into **pre-load-period** variable

.

.

Repeat-cycle

.

.

If **pre-load-period** = 65534 (0xFFFF) then goto SKIP-PRE-LOAD

    If **run-time-ticks** >= **pre-load-period** then

**run-time-ticks** = **run-time-ticks** + **pre-load-value**

**pre-load-period** = 65534 (0xFFFF)

        ; Add pre-load

        ; set pre-load complete

SKIP-PRE-LOAD

.

.

goto Repeat-cycle

**Summary of Command codes (sent from Host)**

| ASCII | HEX code | Description                                                                       | Command sequence (Hex bytes) | Response (Hex bytes)            |
|-------|----------|-----------------------------------------------------------------------------------|------------------------------|---------------------------------|
| M     | 0x4D     | Message1 command, returns null-terminated version message (as ASCII characters)   | 4D 03 4E                     | "SECMFD...IDE... 04/03/19" 0x00 |
| m     | 0x6D     | Message2 command, returns null-terminated Copyright message (as ASCII characters) | 6D 03 6E                     | "IDE Concepts... .." 0x00       |
| P     | 0x50     | Write PIC EEPROM byte value                                                       | 50 xx xx 03 xx               | 80 50 xx 03 xx                  |
| p     | 0x70     | Write PIC EEPROM WORD value                                                       | 70 xx xx xx 03 xx            | 80 70 xx 03 xx                  |
| E     | 0x45     | Read PIC EEPROM byte value                                                        | 45 xx 03 xx                  | 80 45 xx 03 xx                  |
| e     | 0x65     | Read PIC EEPROM WORD value                                                        | 65 xx 03 xx                  | 80 65 xx xx 03 xx               |
| V     | 0x56     | Host Valve ON (returns loaded credit, RFID disabled)                              | 56 xx xx xx xx 03 xx         | 80 56 xx xx xx xx 03 xx         |
| O     | 0x4F     | Host Valve OFF (returns remaining credit and datalog packet)                      | 4F 03 4C                     | 80 4F xx xx xx xx 03 xx         |
| U     | 0x55     | HOST TAP TOP-UP (returns remaining credit and datalog packet)                     | 55 xx xx xx xx 03 xx         | 80 55 xx xx xx xx 03 xx         |
| F     | 0x46     | Factory Reset (restores defaults)                                                 | 46 55 AA 03 BA               | 80 46 03 C5                     |
| K     | 0x4B     | Request to PROGRAM (BOOTLOADER)                                                   | 4B xx xx xx xx 03 xx         | 80 4B xx 03 xx                  |
| R     | 0x52     | Read SPI EEROM DATALOG packet                                                     | 52 xx xx 03 xx               | 80 52 xx xx xx -- -- xx 03 xx   |
| w     | 0x77     | Write DATALOG pointer                                                             | 77 xx xx 03 xx               | 80 77 03 F4                     |
| r     | 0x72     | Read DATALOG pointer                                                              | 72 03 71                     | 80 77 xx xx 03 xx               |
| C     | 0x43     | Set RTC clock / calendar                                                          | 43 xx ----- 03 xx            | 80 43 03 C0                     |
| T     | 0x54     | Get RTC clock / calendar                                                          | 54 03 57                     | 80 54 xx ----- xx 03 xx         |
| L     | 0x4C     | Load Supertap UID + TOP-UP                                                        | 0x4C xx -----xx 03 xx        | 80 4C xx 03 xx                  |
| Z     | 0x5A     | Clear Supertap UID + TOP-UP                                                       | 0x5A xx 03 xx                | 80 5A xx 03 xx                  |
| B     | 0x42     | Bulk Clear Supertap table                                                         | 0x42 03 41                   | 80 42 03 C1                     |
| X     | 0x58     | Get STATUS (returns flags, ADC readings, checks flow)                             | 0x58 03 5B                   | 80 58 xx ----- xx 03 xx         |
| A     | 0x41     | Read TICK-ACCUMULATOR                                                             | 0x41 03 42                   | 80 41 xx -----xx 03 xx          |
| N     | 0x4E     | Write TICK-ACCUMULATOR                                                            | 4E xx ----- 03 xx            | 80 4E 03 CD                     |

**Datalog message (sent from Tap to host system)**

| ASCII | HEX code | Description                              | Command sequence (Hex bytes) | Response (Hex bytes) |
|-------|----------|------------------------------------------|------------------------------|----------------------|
| D     | 0x44     | Automatic DATALOG output (sent from EWC) | 44 xx -----xx 03 xx          | -                    |

**SMARTD consumer message (sent from Tap to SMARTD display)**

| ASCII | HEX code | Description                                      | Command sequence (Hex bytes) | Response (Hex bytes) |
|-------|----------|--------------------------------------------------|------------------------------|----------------------|
| S     | 0x53     | Consumer data message (sent from EWC to display) | 53 xx -----xx 03 xx          | -                    |

**"Version Message" command**

Returns null-terminated firmware message string

<ASCII "M", 0x4D> <ETX> <XOR> (0x4D 0x03 0x4E)

NOTE: no ETX and XOR at end of reply. String is just terminated with a null (00) byte

e.g. "SECMFD\_LP\_EWC2\_BOOT V1.30 30/01/18 IDE eWATERpay Ltd", 0x00

**"Copyright Message" command**

Returns null-terminated firmware message string

<ASCII "m", 0x6D> <ETX> <XOR> (0x6D 0x03 0x6E)

NOTE: no ETX and XOR at end of reply. String is just terminated with a null (00) byte

e.g. "IDE Concepts (IB Technology Ltd), 0x00

**"PIC EEPROM BYTE WRITE" command**

Write byte value to PIC EEPROM

<ASCII "P", 0x50> <ADR> <DATA> <ETX> <XOR> (0x50 xx xx 03 xx)

Where <ADR> = PIC EEPROM address, <DATA> = data byte to write

Reply: 0x80 50 <ADR> 03 <XOR>

0x88 50 <ADR> 03 <XOR> (RS232 error if command format incorrect)

**"PIC EEPROM WORD WRITE" command**

Write WORD (HI Lo bytes) value to PIC EEPROM address <ADR> and <ADR +1>

<ASCII "p", 0x70> <ADR> <DATA\_HI> <DATA\_LO> <ETX> <XOR> (0x50 xx xx xx 03 xx)

Where <ADR> = PIC EEPROM address, <DATA\_HI>, <DATA\_LO> = data bytes (WORD) to write

Reply: 0x80 70 <ADR> 03 <XOR>

0x88 70 <ADR> 03 <XOR> (RS232 error if command format incorrect)

**"PIC EEPROM BYTE READ" command**

Reads and returns byte value from PIC upper "non-sensitive" EEPROM parameter section.

(attempting to read from the lower section containing security parameters will return RS232 ERROR)

<ASCII "E", 0x45> <ADR> <ETX> <XOR> (0x45 xx 0x03 xx)

Reply: 80 45 <ADR> <DATA> 03 <XOR>

Where <ADR> = PIC EEPROM address, <DATA> = Data byte read

0x88 45 <ADR> 03 <XOR> (RS232 error if command format incorrect or invalid address)

**"PIC EEPROM WORD READ" command**

Reads and returns WORD (Hi Lo bytes) value from PIC EEPROM address <ADR> and <ADR +1>

(attempting to read from the lower section containing security parameters will return RS232 ERROR)

<ASCII "e", 0x65> <ADR> <ETX> <XOR> (0x65 xx 0x03 xx)

Reply: 0x80 65 <ADR> <DATA\_HI> <DATA\_LO> 03 <XOR>

Where <ADR> = PIC EEPROM address, <DATA\_HI>, <DATA\_LO> = data bytes (WORD) read

0x88 65 <ADR> 03 <XOR> (RS232 error if command format incorrect or invalid address)

### **"HOST VALVE ON" command**

Turns VALVE ON manually by host control and loads "start credit" value (RFID tag operation is disabled).

<ASCII "V", 0x56> <SCR> <SCR> <SCR> <SCR> <ETX> <XOR>

Where <SCR> <SCR> <SCR> <SCR> = start credit value (32-bit integer, MSB byte first)

Reply: 0x80 <CMD> <SCR> <SCR> <SCR> <SCR> <ETX> <XOR> if OK .

Where <CMD> = 0x56, <ETX> = 0x03

Where <SCR> <SCR> <SCR> <SCR> = start credit loaded (MSB first).

(or 0x88 56 03 DD error reply if VALVE already ON, LOW-FLOW lock-out, LOW-BATT or command format incorrect)

### **"HOST VALVE OFF" command**

Turns VALVE OFF manually by host control and returns end (remaining) credit value.

Datalog packet is also stored and transmitted (with VALVE OFF event code).

RFID operation is enabled afterwards.

<ASCII "O", 0x4F> <ETX> <XOR>

Reply: 0x80 <CMD> <ECR> <ECR> <ECR> <ECR> <ETX> <XOR> if OK

Where <ECR> <ECR> <ECR> <ECR> = end (remaining) credit value (MSB first) and valve OFF.

VALVE\_OFF event datalog packet generated.

0x88 4F 03 C4 error reply if number of arguments, ETX or XOR wrong or If HOST VALVE\_OFF received when RFID enabled.

if end credit value = 0xFF FF FF FF then valve was already OFF (as a result of an error interrupting the water dispensing).

### **"TAP top-up" command**

Adds (tops-up) credit to existing VALVE ON credit during dispensing operation.

Datalog packet is also stored and transmitted (with TOP-UP success event code).

<ASCII "U", 0x55> <SCR> <SCR> <SCR> <SCR> <ETX> <XOR>

Where <SCR> <SCR> <SCR> <SCR> = Top-up credit value (32-bit int, MSB byte first)

Reply: 0x80 <CMD> <ECR> <ECR> <ECR> <ECR> <ETX> <XOR> if OK

Where <ECR> <ECR> <ECR> <ECR> = accumulative (remaining) credit value (MSB first) and TOP-UP event datalog packet generated.

88 55 03 DE error reply if number of arguments, ETX, XOR wrong or TAP TOP-UP received when RFID enabled.

### **"FACTORY RESET" command**

Performs software reset, restores default PIC EEPROM values, sets all DATALOG memory to FFs, clears TICK ACCUMULATOR to zero and clears TOP-UP table. DATALOG pointer reset to 0000. No checks are made for the current state of operation. Typically used to restore EWC to "as new" condition and perform reset.

<ASCII "F", 0x46> <0x55> <0xAA> <ETX> <XOR> (0x46 0x55 0xAA 0x03 0xBA)

Reply: 0x80 46 03 C5, then the EWC performs software reset with default EEPROM parameters loaded.

0x88 46 03 CD (RS232 error if command format incorrect)

NOTE THAT CLEARING THE DATALOG MEMORY TAKES 10MS x 8191 = 80+ SECONDS SO THE GREEN LED IS TOGGLED EVERY 2.5 SECONDS AS A VISUAL INDICATOR BEFORE THE SOFTWARE RESET OCCURS.

### **"REQUEST TO PROGRAM" command**

Sent from host to EWC to check if OK to program, Returns acknowledge + status

Programs EEPROM MAGIC\_BYTE = 0xF0 and THEN jumps to BOOTLOADER code (@0x0000) if OK.

<ASCII "K", 0x4B> <EWC\_ID0, EWC\_ID1, EWC\_ID2, EWC\_ID3> <ETX> <XOR> (4B xx xx xx xx 03 xx)

Where <EWC\_ID0-3> = 4-byte EWC ID (stored in EEPROM) MSB first

Reply: 0x80 4B <STAT> <ETX> <XOR>

Where <STAT> = 0x01 if OK, 0x81 if NOT OK

e.g if EWC\_ID = 00 00 00 01, command sequence = 4B 00 00 00 01 03 49

Reply (0x80 4B 01 03 C9) if OK, Program EEPROM MAGIC\_BYTE = 0xF0 THEN jumps to BOOTLOADER.

Reply (0x80 4B 81 03 49) if NOT OK and operation continues with no action.

If <EWC\_ID0-3> does not match the EWC ID in the EEPROM then the EWC replies with RS232 error:

0x88 4B 03 C0

### **"READ SPI EEPROM LOG" command**

Reads 28-byte SPI EEPROM DATALOG packet from DATALOG NUMBER location

note DATALOG number = absolute address / 32, DATALOG number has range 0 – 8190 (EWC2, 0 – 1022)

<ASCII "R", 0x52> <DATALOG\_NUMBER MSB> <DATALOG\_NUMBER LSB> <ETX> <XOR>

0x52 xx xx 03 xx

Reply: 0x80 52 <DLH> <DLL> xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx xx 03 xx

0x88 52 <DLH> <DLL> 03 <XOR> (RS232 error if command format incorrect)

Where <DLH>, <DLL> = Datalog number MSB, LSB bytes

### **Datalog packet**

EE SS MM HH DD MT YY UU UU UU UU AN RS UC UC SCR SCR SCR SCR ECR ECR ECR ECR FC FC FC FT FT  
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Where EE = Error/Event code

SS MM HH = Seconds, Minutes, Hours bytes (BCD), DD MT YY = Day, Month, Year bytes (BCD)

UU UU UU UU is the 4-byte card UID (MSB -> LSB)

AN = Battery voltage ADC value (calibration: voltage = ADC value/256 x 15, for example 12v = 0xCC)

RS = Reserved for future use (default 0x00)

UC UC is the 2-byte usage counter

SCR SCR SCR SCR is the 4-byte start Credit value (MSB -> LSB) stored in MilliCredit units (MITs)

ECR ECR ECR ECR is the 4-byte end Credit value

FC FC FC is the 3-byte (MSB, MID, LSB) Flow meter count (flow volume). Litres ~ = FC / 360

FT FT is the 2-byte (MSB, LSB) Flow time (seconds)

### **"WRITE DATALOG POINTER" command**

Stores 2-byte DATALOG number in RAM (and non-volatile store)

<ASCII "w", 0x77> <DATALOG\_NUMBER MSB> <DATALOG\_NUMBER LSB> <ETX> <XOR> (0x77 xx xx 03 xx)

Reply: 0x80 77 03 F4

0x88 77 03 FC (RS232 error if command format incorrect)



### **"READ DATALOG POINTER" command**

Returns two bytes DATALOG number from RAM store

<ASCII "r", 0x72> <ETX> <XOR> (0x72 0x03 0x71)

Reply: 0x80 72 <DLH> <DLL> 03 <XOR>

0x88 72 03 F9 (RS232 error if command format incorrect)

Where <DLH>, <DLL> = Datalog number MSB, LSB bytes

NOTE: DATALOG number is stored in non-volatile memory and retained on EWC reset/power-on. FACTORY RESET clears DATALOG pointer to 0000.

### **"SET CLOCK" command**

Sets main time/calendar registers of RTC chip. Note: assumes the BCD values and format are correct, no checks for validity are made.

<ASCII "C", 0x43> <SS> <MM> <HH> <DD> <MN> <YY> <ETX> <XOR>

Example: 0x43 30 50 10 09 02 16 03 2D sets the RTC to 10:50:30 time and 09/02/2016 date

Reply: 0x80 43 03 C0

0x88 43 03 C8 (RS232 error if command format incorrect)

### **"GET TIME" command**

Reads and returns the PCF 2123 RTC time/calendar registers values

<ASCII "T", 0x54> <ETX> <XOR> (0x54 0x03 0x57)

Reply: 0x80 54 <SS> <MM> <HH> <DD> <MN> <YY> 03 <XOR>

Where <SS>....<YY> are BCD values.

0x88 54 03 DF (RS232 error if command format incorrect)

NOTE that most significant bit (b7) of the seconds BCD value is set if the RTC oscillator has been interrupted (Reset/power-on has occurred for example).

The EWC2 loads default time and date if "seconds" BCD b7 = 1, 12:00 o'clock, 01/01/2015 and clears the seconds registers so the host should never see this condition.

### **"LOAD SUPERTAP TOP-UP" command**

Loads user UID and TOP-UP credit values into EWC EEPROM memory at specified table location 0-24.

<ASCII "L", 0x4C> <LOC> <UID> <UID> <UID> <UID> <TOP-UP> <TOP-UP> <TOP-UP> <TOP-UP> <ETX> <XOR>

Where <LOC> = table location 0-24 (as hex byte)

<UID0> <UID1> <UID2> <UID3> = plain/encrypted tag UID (stored MSB first)

<TOP-UP0> <TOP-UP1> <TOP-UP2> <TOP-UP3> = plain/encrypted 32-bit integer "top-up" to add or write to tag (MSB first)

Most Significant bit of <TOP-UP0> indicates WRITE credit (b31 = 1) or ADD credit (b31 = 0)

Reply: 0x80 4C <LOC> 03 <XOR> (or 0x88 4C <LOC> 03 <XOR> error reply if table location is out of range, or command format incorrect)

### **"CLEAR SUPERTAP TOP-UP" command**

Clears user UID and TOP-UP credit values in EWC EEPROM memory at specified table location 0-24.

<ASCII "Z", 0x5A> <LOC> <ETX> <XOR> (0x5A xx 03 xx)

Where <LOC> = table location 0-24 (as hex byte)

Reply: 0x80 5A <LOC> 03 <XOR>

### **"BULK CLEAR SUPERTAP table" command**

Clears entire 25 x 8-byte (200-byte) SuperTap EEPROM table.

<ASCII "B", 0x42> <ETX> <XOR> (0x42 03 41)

Reply: 0x 80 42 03 C1

(or 0x88 42 03 C9 error reply if table location is out of range, or number of arguments wrong)

(or 0x81 42 03 C0 error reply indicating **Internal EEPROM error** (fatal h/w error and very unlikely))

### **"GET STATUS" command**

Generates STATUS message containing system flags, ADC readings and flow information.

<ASCII "X", 0x58> <ETX> <XOR> (0x58 03 5B)

Reply: 0x88 58 03 8B (RS232 error if command format incorrect)

or 0x80 58 <22-byte STATUS data > 03 <XOR> if command received and handled OK.

Typical status reply:

```
0x80 58 <SS MM HH DD MN YY UU UU UU UU VBAT VWAT RS2 RS3 RS4 RS0 RS1 FLG0 FLG1 FC0 FC1 ...
      0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
                                     ... FT> 03 <XOR>
                                           21 22 23
```

Where:

UU UU UU UU = tag UID 0-3 bytes (if tag present)

VBAT = Battery voltage ADC value (calibration: voltage = ADC value/256 x 15, for example 12v = 0xCC)

VWAT = VWAT ADC value (measures resistive loads such as pressure switch)

RS0, RS1, RS2, RS3, RS4 = reserved bytes for future use

FLG0 = system flag bits.

FLG1 = reserved byte for future use

FC0, FC1 = Sample flow count HI, LO bytes for 1-second period

FT = Sample period (x20ms)

### **FLG0 bit allocations**

b7 b6 b5 b4 b3 b2 b1 b0

| | | | | | | TAMP2 (Tamper 2 switch, bottom box. NOTE: NOT DEFINED ON EWC V2.3)

| | | | | | | TAMP1 (Tamper 1 switch, on solar4 board)

| | | | | | | GSM\_FLG (1 = GSM not-locked, from Gadwall RTS)

| | | | | | | VALVE ON/OFF status (1 = ON, 0 = OFF)

| | | | | | | LOCK\_FLG (1 = no flow lockout timeout in progress)

| | | | | | | PROX\_FLG (1 = Magnet detected but no RFID, event message sent/logged)

| | | | | | | LBATT\_FLG (1 = low battery detected, event message sent/logged)

| | | | | | | RFID\_FLG (1 = disabled, manual valve control)

### TICK-ACCUMULATOR commands

8-byte TICK ACCUMULATOR store added to external (non-volatile) SPI EEPROM.

The tick accumulator is used to maintain a "total" tick (flow meter pulses) count record and is effectively an 8-byte (64-bit) non-volatile integer. The tick accumulator is updated each time a datalog packet with associated FC (flow count) is stored/transmitted.

$TICK\_ACCUMULATOR = TICK\_ACCUMULATOR + FC$

**READ\_TICK\_ACC** and **WRITE\_TICK\_ACC** commands provide the means to read and write (or clear) the 8-byte tick accumulator value. Maximum tick accumulator value is  $2e64 = 1.84 \times 10e19$  ticks =  $5.59 \times 10e16$  litres (based on 360 ticks per litre conversion).

### "READ TICK ACC" command

<ASCII "A",0x41> <ETX> <XOR>

Example: 0x41 03 42 (Return 8-byte tick accumulator integer).

Reply: 88 41 03 CA (RS232 error if command format incorrect)

or 80 41 <8-byte TICK ACCUMULATOR> 03 <XOR> if command received and handled OK.

### "WRITE TICK ACC" command

<ASCII "N",0x4E> < 8-byte Tick Accumulator integer > <ETX> <XOR>

Example: 0x4E 00 00 00 00 00 00 00 03 4D (Clear 8-byte tick accumulator).

Reply: 88 4E 03 C5 (RS232 error if command format incorrect)

or 80 4E 03 CD if command received and handled OK.

### Summary of program/reset start-up operation:

1) A FULL programming operation of the EWC (using "BOOT" version of firmware) will restore all EEPROM parameters to default values including the EWC ID, power-up counter and all the TOP-UP table contents (zeros written for all 25 x UID and top-up values) and set the DATALOG pointer to zero.

2) An OTA (Over The Air) programming operation of the EWC (using OTA version of firmware) does not include the BOOTLOADER code section and will restore EEPROM parameters to default values and set the DATALOG pointer to zero. BUT THE EWC ID, POWER-UP COUNTER, AND THE TOP-UP TABLE CONTENTS ARE NOT CHANGED.

3) A FACTORY RESET command will set all the DATALOG memory to "FF" values as if a "new" device, clear the TICK\_ACCUMULATOR, TOP-UP table and DATALOG pointer to zero. A software reset is then performed.

Note that the FULL/OTA programming operations DO NOT initialise or clear the DATALOG SPI EEPROM or Tick Accumulator, only sending a FACTORY RESET command will do this.

**EWC EEPROM memory contains configuration parameters as below:**

Polling delay parameter. Time

|       |            |
|-------|------------|
| 0x00, | 4.096 mS   |
| 0x10, | 8.192 mS   |
| 0x20, | 16.384 mS  |
| 0x30, | 32.768 mS  |
| 0x40, | 65.536 mS  |
| 0x50, | 131.072 mS |
| 0x60, | 262.144 mS |
| 0x70, | 524.288 mS |
| 0x80, | 1.048 secs |
| 0x90, | 2.097 secs |
| 0xA0, | 4.194 secs |
| 0xB0, | 8.388 secs |

**EEPROM Byte address**
**Default value**

Byte 0: Reserved (not used)

H'00'

Byte 1: Checksum Value (Reserved – DO NOT CHANGE)

H'FF'

Byte 2: EWC Block Read address on card (Mifare card block address 0-255)

H'0A'

Byte 3: Key number/type to access EWC data

H'05'

(TxxKKKKK), (T = Key type, 0 = KeyA, 1 = KeyB)

(K = Key code number, 0 - 31)

Byte 4: Encryption ON/OFF control (Default ON = 0xFF, OFF = 0x00)

H'FF'

Byte 5: ) 32 bit Encryption Seed1 (M.S byte)

H'01'

Byte 6: )

H'03'

Byte 7: )

H'09'

Byte 8: ) (L.S byte)

H'07'

6-byte keycode (KEYA) for EWC block access

Byte 9: ) KEYA\_0

H'EF'

Byte 10: ) KEYA\_1

H'F2'

Byte 11: ) KEYA\_2

H'6D'

Byte 12: ) KEYA\_3

H'53'

Byte 13: ) KEYA\_4

H'42'

Byte 14: ) KEYA\_5

H'8C'

4-byte EWC authentication code on card

Byte 15: ) AUTH\_0 (MSB)

H'8F'

Byte 16: ) AUTH\_1

H'67'

Byte 17: ) AUTH\_2

H'CE'

Byte 18: ) AUTH\_3 (LSB)

H'B9'

Byte 19 (0x13): SuperT 32 bit Encryption mask (Pseudo UID) (M.S byte)

H'E9'

Byte 20 (0x14):

H'FD'

Byte 21 (0x15):

H'1A'

Byte 22 (0x16): (L.S byte)

H'B7'

↓ Non security related EEPROM parameters, Readable by READ PIC EEPROM command

|                                                                                                                                                                              |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Byte 23: (0x17): PRE-LOAD-VALUE ticks (MSB)                                                                                                                                  | H'01' |
| Byte 24: (0x18): PRE-LOAD-VALUE (LSB) Default = 360 dec = 0x0168                                                                                                             | H'68' |
| Byte 25 (0x19): FCF0 Flow conversion factor, ticks per credit DIVISOR (MSB)                                                                                                  | H'01' |
| Byte 26 (0x1A): FCF1 DIVISOR (LSB), (default = 491 decimal, 0x01EB)                                                                                                          | H'EB' |
| Byte 27 (0x1B): HEALTH_STATE report period<br>(OFF = 00 or 1 – 255 minute intervals, default OFF)                                                                            | H'00' |
| Byte 28 (0x1C): No-flow cycle count timeout value (Default = 14-seconds)                                                                                                     | H'0E' |
| Byte 29 (0x1D): NO-FLOW error low flow (pulse) count MSB                                                                                                                     | H'01' |
| Byte 30 (0x1E): LSB (Default = 1 litre = 360 dec = 0x0168)                                                                                                                   | H'68' |
| Byte 31: (0x1F): Valve Drive ON/OFF timeout (default = 6 secs)<br>(zero setting selects 30mS pulse for latching solenoid valves)                                             | H'06' |
| Byte 32: (0x20): No Flow error LOCK-OUT timeout (default = 30 secs)                                                                                                          | H'1E' |
| Byte 33: (0x21): Tag Polling period 1 HI byte (= seconds x 256)<br>(default 1-hour, 3600 seconds = 0x0E10)                                                                   | H'0E' |
| Byte 34: (0x22): Tag Polling period 2 HI byte (= seconds x 256)<br>(default 2-hours, 7200 seconds = 0x1C20)                                                                  | H'1C' |
| Byte 35: (0x23): SMARTD message control (Default ON = 0xFF , OFF = 00)                                                                                                       | H'FF' |
| Byte 36: (0x24): Magnetic Prox control (Default ON = 0xFF, OFF = 0x00)                                                                                                       | H'FF' |
| Byte 37: (0x25): NO_FLOW error control (Default ON = 0xFF, OFF = 0x00)                                                                                                       | H'FF' |
| Byte 38: (0x26): PRE-LOAD-PERIOD (ticks MSB)                                                                                                                                 | H'00' |
| Byte 39: (0x27): PRE-LOAD-PERIOD (LSB) Default = 0, start of dispense period<br>(When Dispensing flow-count >= PRE-LOAD-PERIOD,<br>flow-count = flow-count + PRE-LOAD-VALUE) | H'00' |
| Byte 40: (0x28): Low VBATT limit (default OFF point, 11.9v = 203, 0xCB)                                                                                                      | H'CB' |
| Byte 41: (0x29): High VBATT limit (default ON point, 12.4v = 212 = 0xD4)                                                                                                     | H'D4' |
| Byte 42: (0x2A): Dispense "time" limit HI byte (= seconds x 256)<br>(default > 0x0708, 1800 secs = 30 mins)                                                                  | H'07' |
| Byte 43: (0x2B): Dispense "flow count limit" HI byte ( = flow count x 65536)<br>(default > 0x010000, 65536 / 360 ~ = 180 Litres)                                             | H'01' |
| Byte 44: (0x2C): FX0, Currency conversion factor MSB.                                                                                                                        | H'00' |
| Byte 45: (0x2D): FX1 (Price/Credit as 1/1,000,000 of unit credit)                                                                                                            | H'00' |
| Byte 46: (0x2E): FX2 (default UK price = 0.00111 p/L x 10^6 = 1110 = 0x0456)                                                                                                 | H'04' |
| Byte 47: (0x2F): FX3, LSB                                                                                                                                                    | H'56' |
| Byte 48: (0x30): LCF0, Litres Conversion Factor, ticks per litre (MSB)                                                                                                       | H'01' |
| Byte 49: (0x31): LCF1 (LSB) (default = 360 decimal, 0x0168)                                                                                                                  | H'68' |

↓ EEPROM parameters not affected by Factory Reset “reload”

|                                                                    |                                  |       |
|--------------------------------------------------------------------|----------------------------------|-------|
| Byte 50 (0x32): EWC_ID (MSB)                                       | 4-byte ID code for the EWC board | H'00' |
| Byte 51 (0x33): EWC_ID                                             |                                  | H'00' |
| Byte 52 (0x34): EWC_ID                                             |                                  | H'00' |
| Byte 53 (0x35): EWC_ID (LSB)                                       | default ID = 00 00 00 02         | H'02' |
|                                                                    |                                  |       |
| Byte 54 (0x36): Power-up count (increments on each power-on/reset) |                                  | H'00' |
| Byte 55 (0x37): BOOTLOADER MAGIC BYTE (DO NOT CHANGE)              |                                  | H'FF' |

↓ SuperTap Table (25 x 8-byte entries)

|                               |                   |       |
|-------------------------------|-------------------|-------|
| Byte 56 (0x38): UID0 (MSB)    |                   | H'00' |
| Byte 57 (0x39): UID0          | SuperT Location 0 | H'00' |
| Byte 58 (0x3A): UID0          |                   | H'00' |
| Byte 59 (0x3B): UID0 (LSB)    |                   | H'00' |
|                               |                   |       |
| Byte 60 (0x3C): TOP_UP0 (MSB) |                   | H'00' |
| Byte 61 (0x3D): TOP_UP0       |                   | H'00' |
| Byte 62 (0x3E): TOP_UP0       |                   | H'00' |
| Byte 63 (0x3F): TOP_UP0 (LSB) |                   | H'00' |
|                               |                   |       |
| Byte 64 (0x40): UID1 (MSB)    |                   | H'00' |
| Byte 65 (0x41): UID1          | SuperT Location 1 | H'00' |
| Byte 66 (0x42): UID1          |                   | H'00' |
| Byte 67 (0x43): UID1 (LSB)    |                   | H'00' |
|                               |                   |       |
| Byte 68 (0x44): TOP_UP1 (MSB) |                   | H'00' |
| Byte 69 (0x45): TOP_UP1       |                   | H'00' |
| Byte 70 (0x46): TOP_UP1       |                   | H'00' |
| Byte 71 (0x47): TOP_UP1 (LSB) |                   | H'00' |

↓

|                 |                   |
|-----------------|-------------------|
| Byte 72 (0x48)  | SuperT Location 2 |
| Byte 80 (0x50)  | SuperT Location 3 |
| Byte 88 (0x58)  | SuperT Location 4 |
|                 |                   |
| Byte 96 (0x60)  | SuperT Location 5 |
| Byte 104 (0x68) | SuperT Location 6 |
| Byte 112 (0x70) | SuperT Location 7 |
| Byte 120 (0x78) | SuperT Location 8 |
| Byte 128 (0x80) | SuperT Location 9 |



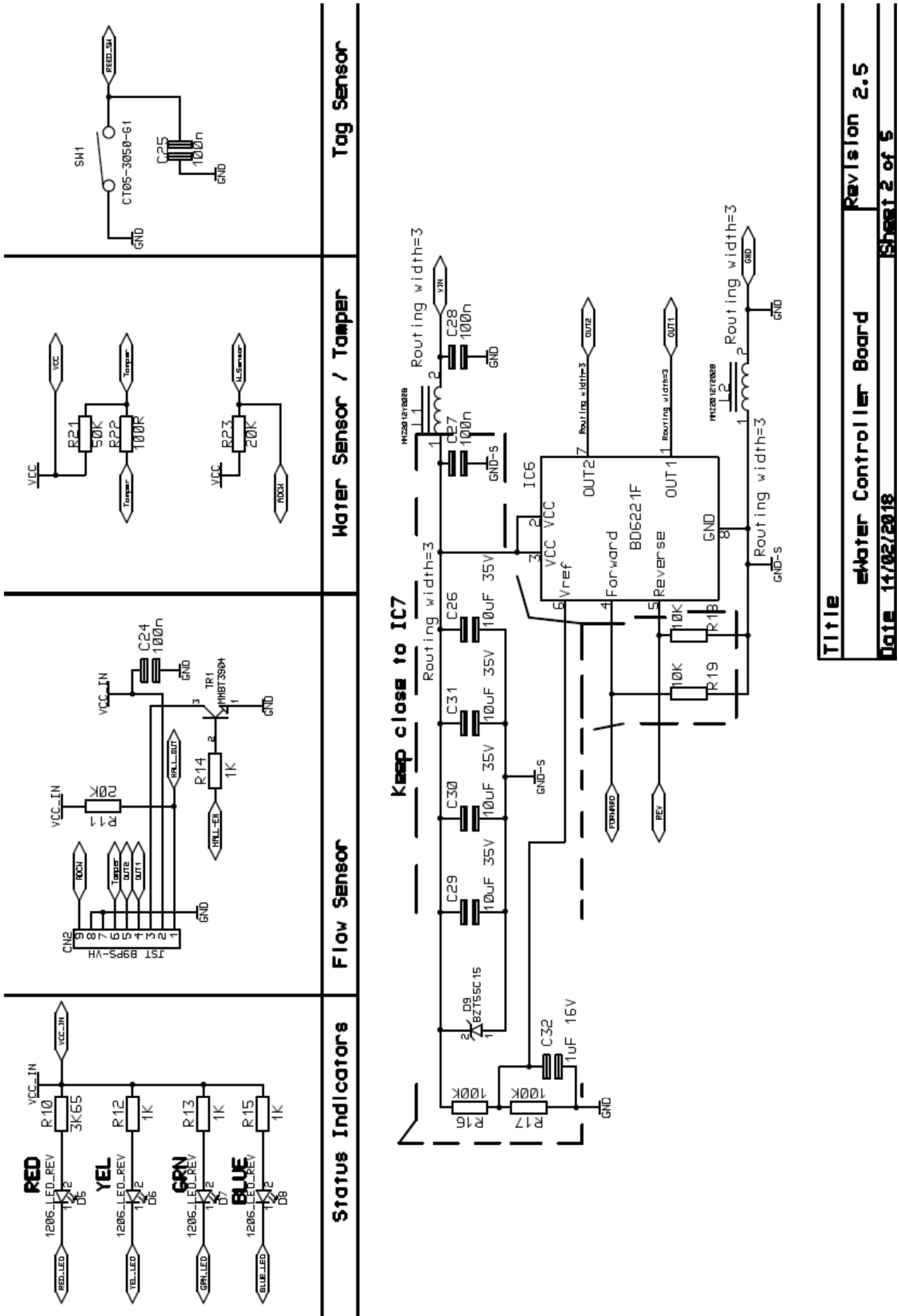
|                 |                    |
|-----------------|--------------------|
| Byte 136 (0x88) | SuperT Location 10 |
| Byte 144 (0x90) | SuperT Location 11 |
| Byte 152 (0x98) | SuperT Location 12 |
| Byte 160 (0xA0) | SuperT Location 13 |
| Byte 168 (0xA8) | SuperT Location 14 |
| Byte 176 (0xB0) | SuperT Location 15 |
| Byte 184 (0xB8) | SuperT Location 16 |
| Byte 192 (0xC0) | SuperT Location 17 |
| Byte 200 (0xC8) | SuperT Location 18 |
| Byte 208 (0xD0) | SuperT Location 19 |
| Byte 216 (0xD8) | SuperT Location 20 |
| Byte 224 (0xE0) | SuperT Location 21 |
| Byte 232 (0xE8) | SuperT Location 22 |
| Byte 240 (0xF0) | SuperT Location 23 |



|                                 |                    |       |
|---------------------------------|--------------------|-------|
| Byte 248 (0xF8): UID24 (MSB)    |                    | H'00' |
| Byte 249 (0xF9): UID24          | SuperT Location 24 | H'00' |
| Byte 250 (0xFA): UID24          |                    | H'00' |
| Byte 251 (0xFB): UID24 (LSB)    |                    | H'00' |
| Byte 252 (0xFC): TOP_UP24 (MSB) |                    | H'00' |
| Byte 253 (0xFD): TOP_UP24       |                    | H'00' |
| Byte 254 (0xFE): TOP_UP24       |                    | H'00' |
| Byte 255 (0xFF): TOP-UP24 (LSB) |                    | H'00' |

~~~~~  
End of EEPROM





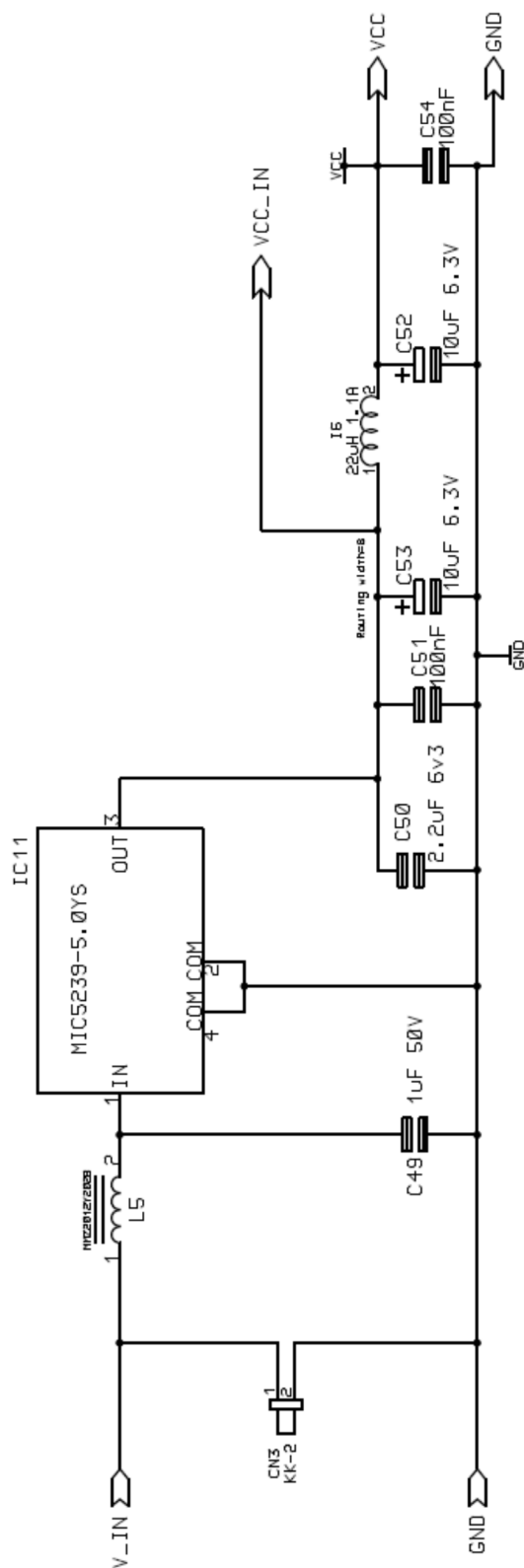
Title

eWater Controller Board

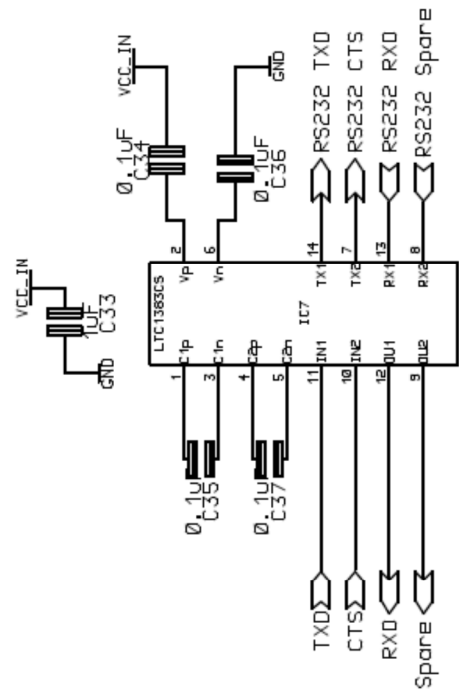
Revision 2.5

Date 14/02/2018

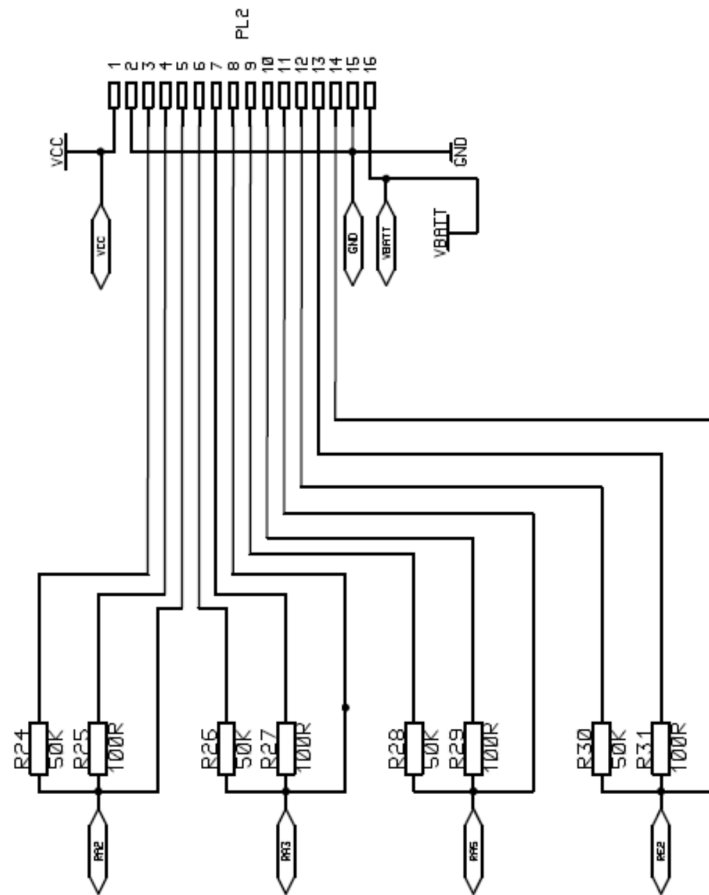
Sheet 2 of 5



|                        |            |              |
|------------------------|------------|--------------|
| Title                  |            | Revision     |
| Water Controller Board |            | 2.5          |
| Date                   | 14/02/2018 | Sheet 3 of 5 |



|                         |              |
|-------------------------|--------------|
| Title                   |              |
| eWater Controller Board | Revision 2.5 |
| Date 14/02/2018         | Sheet 4 of 5 |



|                        |                     |
|------------------------|---------------------|
| <b>Title</b>           | <b>Revision</b>     |
| Water Controller Board | 2.5                 |
| <b>Date</b> 14/02/2018 | <b>Sheet</b> 5 of 5 |



## EWC 2.5 Bill of Materials 14/08/20

EWC 2.5 Bill of Materials 14/08/20 (Note: C17 variable tuning capacitor and C20, C38 fixed capacitors removed. C7, C8 replaced with 180pF 1% tolerance NPO/COG capacitor types to give best compromise tuning for laminated and molded tags with embedded N52 magnets)  
NOTE: connecting pin header now fitted to CN1 and CN3 removed. Soft-iron disc thickness increased to 3.5mm

| Location                                          | Part Number                                  | Description                                                                                                                                                                          | Package       | QTY Per PCB | RS         | Farnell   |
|---------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------|------------|-----------|
| BAT1                                              | SEIKO MS621FE-FL11E                          | Rechargeable Battery, Coin Cell, Single Cell, 3 V, Lithium Manganese Dioxide, 5.5 mAh, Button Cell                                                                                   | MS621FE-FL11E | 1           | 161-4634   |           |
| D1                                                | S2A                                          | S2A                                                                                                                                                                                  | SMB           | 1           |            | 1467485   |
| IC4                                               | MicroChip AT25M02-SSH                        | Serial-SPI EEPROM Memory IC 2Mb (256K x 8) SPI 5MHz 8-SOIC                                                                                                                           | SO8           | 1           |            | Avnet     |
| XT2                                               | Abracon ABS09-32.768KHZ-7-T                  | ABRACON ABS09-32.768KHZ-7-T CRYSTAL, 32.768KHZ, 7PF, 4.1 X 1.5MM                                                                                                                     | ABS09-7-T     | 1           |            | 2467867   |
| D2                                                | NXP BAT721.215                               | NXP BAT721.215 Small Signal Schottky Diode, Single, 40 V, 200 mA, 550 mV, 1 A                                                                                                        | SOT23         | 1           |            | 1757754RL |
| IC6                                               | ROHM BD6221F-E2                              | ROHM BD6221F-E2 IC, H BRIDGE DRIVER, 18V, 1A, SOP8                                                                                                                                   | SO8           | 1           |            | 6996746   |
| XT1                                               | AKER - C3E-13.560-12-1010-X,13.560MHZ        | CRYSTAL, 2.5X3MM, CER, 13.560MHZ                                                                                                                                                     | C3EXTAL       | 1           |            | 1640977   |
| C1, C4-C6, C10, C18, C23, C24, C27, C28, C51, C54 | TDK C1608X5R1H104K80AA                       | CAP, MLCC, X5R, 100NF, 50V, 0603, 10% (Note C25 is NOT FITTED)                                                                                                                       | 0603R         | 14          |            | 2211177   |
| C19                                               | AVX 06035A101JAT2A                           | AVX 06035A101JAT2A SMD Multilayer Ceramic Capacitor, 0603 [1608 Metric], 100 pF, 50 V, ± 5%, COG / NPO                                                                               | 0603R         | 1           |            | 499122RL  |
| C7, C8                                            | Kemet C0603C181F5GACTU                       | <b>C0603C181F5GACTU SMD Multilayer Ceramic Capacitor, 180 pF, 50 V, 0603 [1608 Metric], ± 1%, COG / NPO (NOTE 1% tolerance)</b>                                                      | 0603R         | 2           |            | 2820831   |
| C2, C3                                            | MULTICOMP MC0603N100D500CT                   | CHPC0603, 10pF MULTICOMP MC0603N100D500CT SMD Multilayer Ceramic Capacitor, MC Series, 10 pF, ± 0.5pF, COG / NPO, 50 V, 0603                                                         | 0603R         | 2           |            | 1759053RL |
| C16                                               | MURATA - GRM1885C1H150JA01D                  | CAP, MLCC, COG/NPO, 15PF, 50V, 0603, 5%                                                                                                                                              | 0603R         | 1           |            | 8819823   |
| C20                                               | Not fitted                                   | (NOT FITTED)                                                                                                                                                                         | 0603R         | n/a         |            | n/a       |
| C38                                               | Not fitted                                   | (NOT FITTED)                                                                                                                                                                         | 0603R         | n/a         |            | n/a       |
| C21, C32, C49                                     | TDK TDK C1608X5R1H105K080AB                  | TDK C1608X5R1H105K080AB SMD Multilayer Ceramic Capacitor, C Series, 1 µF, ± 10%, X5R, 50 V, 0603                                                                                     | 0603R         | 3           |            | 2211179RL |
| C33-C37                                           | GRM188R71H104KA93D                           | Murata 0603 50V X7R 0.1uF 10% 50V                                                                                                                                                    | 0603R         | 5           |            | 8820023   |
| C50                                               | MURATA - GRM1885C1H220JA01D                  | MULTICOMP MC0603B225K6R3CT SMD Multilayer Ceramic Capacitor, MC Series, 2.2 µF, ± 10%, X7R, 6.3 V, 0603                                                                              | 0603R         | 1           |            | 2320810RL |
| C9, C13                                           | MURATA - GRM1885C1H220JA01D                  | CAP, MLCC, COG/NPO, 22PF, 50V, 0603, 5%                                                                                                                                              | 0603R         | 2           |            | 8819831   |
| C11, C12, C14, C15                                | MURATA - GRM1885C1H680JA01D                  | CAP, MLCC, COG/NPO, 68PF, 50V, 0603, 5%                                                                                                                                              | 0603R         | 4           |            | 8819858   |
| C26, C29-C31                                      | Murata GRM31CR6YA106KA12L                    | SMD Multilayer Ceramic Capacitor, GRM Series, 10 µF, ± 10%, X5R, 35 V, 1206                                                                                                          | 1206R         | 4           |            | 1797011RL |
| R3, R4                                            | EUROHM - RY000 or Equivalent                 | 0R, Thick Film, 100mW, 1%, 50V, 0603                                                                                                                                                 | 0603R         | 2           |            |           |
| R16                                               | EUROHM - RB104 or Equivalent                 | 100K, Thick Film, 100mW, 1%, 50V, 0603                                                                                                                                               | 0603R         | 1           |            |           |
| R2,R8,R22,R25,R27,R29,                            |                                              |                                                                                                                                                                                      |               |             |            |           |
| R31                                               | EUROHM - RB101 or Equivalent                 | 100R, Thick Film, 100mW, 1%, 50V, 0603                                                                                                                                               | 0603R         | 7           |            |           |
| R6, R18, R19                                      | EUROHM - RB103 or Equivalent                 | 10K, Thick Film, 100mW, 1%, 50V, 0603                                                                                                                                                | 0603R         | 3           |            |           |
| R12, R13, R14,R15                                 | EUROHM - RB102 or Equivalent                 | 1K, Thick Film, 100mW, 1%, 50V, 0603                                                                                                                                                 | 0603R         | 4           |            |           |
| R7                                                | EUROHM - RB272 or Equivalent                 | 2K7, Thick Film, 100mW, 1%, 50V, 0603                                                                                                                                                | 0603R         | 1           |            |           |
| R10                                               | EUROHM - RB362 or Equivalent                 | 3K6, Thick Film, 100mW, 1%, 50V, 0603                                                                                                                                                | 0603R         | 1           |            |           |
| R9                                                | EUROHM - RB821 or Equivalent                 | 820R, Thick Film, 100mW, 1%, 50V, 0603                                                                                                                                               | 0603R         | 1           |            |           |
| R20                                               | EUROHM - RB153 or Equivalent                 | 15K, Thick Film, 100mW, 1%, 50V, 0603                                                                                                                                                | 0603R         | 1           |            |           |
| R5, R1,R11,R23                                    | EUROHM - RB203 or Equivalent                 | 20K, Thick Film, 100mW, 1%, 50V, 0603                                                                                                                                                | 0603R         | 4           |            |           |
| R21,R24,R26,R28,R30                               | EUROHM - RB513 or Equivalent                 | 51K, Thick Film, 100mW, 1%, 50V, 0603                                                                                                                                                | 0603R         | 5           |            |           |
| C52, C53                                          | AVX TAJA106K006RNJ                           | AVX TAJA106K006RNJ CAP, TANT, 10UF, 6.3V, CASE A                                                                                                                                     | C-TAJ-A       | 2           |            | 197014    |
| IC2                                               | NXP MFRCS3101T/0FE,112 or CLRC63201T/0FE,112 | MFRCS3101T/0FE,112 (alternative CLRC63201T/0FE,112)                                                                                                                                  | SO32          | 1           |            |           |
| SW1                                               | CT05-3050-G1                                 | COTO TECHNOLOGY CT05-3050-G1 Reed Switch, SPST-NO, SMD, 500 mA, 140 VDC 5W                                                                                                           | CT05-3050-G1  | 1           |            | 1081694RL |
| D9                                                | BZT55C15                                     | Zener Single Diode, BZT55C Series, 15 V, 500 mW, SOD-80 (MiniMELF), 2 Pins, 200 °C                                                                                                   | Melf          | 1           |            | 1621844   |
| D3                                                | Diode LL4148                                 | Small Signal Diode, Single, 100 V, 200 mA, 1 V, 4 ns, 2 A (low reverse leakage current)                                                                                              | LL4148        | 1           |            | 2101203   |
| I1, I2                                            | WURTH ELEKTRONIK 74476010C                   | WURTH ELEKTRONIK 74476010C, WE-KI Series, 10 nH, 600 mA, 0805, Wirewound                                                                                                             | 805           | 2           |            | 1748800   |
| I6                                                | Bourns CM453232-220KL                        | BOURNS - CM453232-220KL, 22.0UH                                                                                                                                                      | IND0481-C     | 1           |            | 9358080   |
| I3, I4                                            | NLV32T-1R0J-PF                               | TDK NLV32T-1R0J-PF Surface Mount High Frequency Inductor, NLV Series, 1 µH, ± 5%, 1210                                                                                               | 1210R         | 2           |            | 9621490RL |
| CN1                                               | Molex 22 11 2502                             | Molex 2112052 KK254 6373 5-way, 2.54mm pitch, friction lock (male) pin header or equivalent                                                                                          |               | 1           |            | 2751316   |
| CN2                                               | JST B9PS-VH                                  | JST B9PS-VH(LF)(SN) - Wire-To-Board Connector, 3.96 mm, 9 Contacts, Header, VH Series, Through Hole, 1 Rows                                                                          | JST B9PS-VH   | 1           |            | 2399170   |
| CN3                                               | TAICOM TK802MSL (NOT FITTED)                 | 2.54MM WAFER 2 WAY + LOCK (NOT FITTED)                                                                                                                                               | KK-2R         | n/a         |            |           |
| D5                                                | WURTH ELEKTRONIK 156 120 R57 530 0           | WL-SMRW Reverse mount mono-color round 1206, Red Typ.(@20mA): 624nm, 60mcd, 2.0V, 140° AlnGaP                                                                                        | 1206_LED_REV  | 1           |            | 2322094   |
| D6                                                | WURTH ELEKTRONIK 156 120 Y57 530 0           | WL-SMRW Reverse mount mono-color round 1206, Yellow, Typ.(@20mA): 589nm, 80mcd, 2.0V, 140° AlnGaP                                                                                    | 1206_LED_REV  | 1           |            | 2322098   |
| D7                                                | WURTH ELEKTRONIK 156 120 G57 530 0           | WL-SMRW Reverse mount mono-color round 1206, Green, Typ.(@20mA): 525nm, 235mcd, 3.3V, 140° InGaN                                                                                     | 1206_LED_REV  | 1           |            | 2322091   |
| D8                                                | WURTH ELEKTRONIK 156 120 B57 530 0           | WL-SMRW Reverse mount mono-color round 1206, Blue, Typ.(@20mA): 470nm, 50mcd, 3.2V, 140° nGaN                                                                                        | 1206_LED_REV  | 1           |            |           |
| IC7                                               | Linear Tech LTC1383CS                        | transceiver RS232, 2 Drivers, 2 Receivers, 5V supply, SOIC-16                                                                                                                        | SO16          | 1           |            |           |
| TR1                                               | NXP MMBT3904.215                             | NXP MMBT3904.215 Bipolar (BJT) Single Transistor, Switching, NPN, 40 V, 300 MHz, 250 mW, 200 mA, 100 Hfe                                                                             | SOT-23W       | 1           |            | 1757935RL |
| L1, L2, L5                                        | WURTH ELEKTRONIK 742 792 208 00              | WE-MPSB EMI Multilayer Power Suppression Bead, 0805, 4Amps                                                                                                                           | 0805R         | 3           |            |           |
| IC5                                               | NXP PCF2123TS/1.118                          | NXP PCF2123TS/1.118 Real Time Clock                                                                                                                                                  | TS50P14       | 1           | RS-7275731 |           |
| IC1                                               | MICROCHIP PIC16F887T-I/PT                    | MICROCHIP PIC16F887T-I/PT 8 Bit Microcontroller, Flash, 20 MHz, 14 KB, 368 Byte, 44 Pins, TQFP                                                                                       | TQFP-44       | 1           |            | 2315832   |
| P1                                                | WURTH ELEKTRONIK 61301021121                 | Board-To-Board Connector, Dual, 2.54 mm, 10 Contacts, straight Header, WR-PHD Series, Through Hole, 2 Rows                                                                           | IDC10         | 1           |            | 1841232   |
| C17                                               | Murata - TZB4R500AA10R00                     | Ceramic 7/50pF 50V dc SMD Trimmer (NOT FITTED)                                                                                                                                       | TRIM-CAP-CTZ3 | n/a         |            | n/a       |
| IC11                                              | MicroChip MIC5239-5.0YS                      | Fixed LDO Voltage Regulator, 2.3V to 30V, 350mV Dropout, 5Vout, 500mAout, SOT-223-3                                                                                                  | SOT-223       | 1           |            | 1494947   |
| PL2                                               | WURTH ELEKTRONIK 61201621721                 | Wire-To-Board Connector, Right Angle, 2.54 mm, 16 Contacts, Header, WR-BHD Series, Through Hole                                                                                      |               | 1           |            | 1642024   |
| PCB                                               | P18-04                                       | P18-04 1.6mm FR4 material, 1oz Cu weight. Note recessed hole (12mm diameter) for soft iron disc milled to half of PCB thickness (0.8mm depth from non-component/bottom layer of PCB) | PCB           | 1           |            |           |
| METAL-DISC                                        | Soft iron disc                               | Soft iron disc 12mm diameter x 3.5 mm thick. Glued into recess in bottom layer of PCB (glue must be water resistant)                                                                 | DISC          | 1           |            |           |